

Clamper

peak to peak

20

10

10

5

20

15

10

5

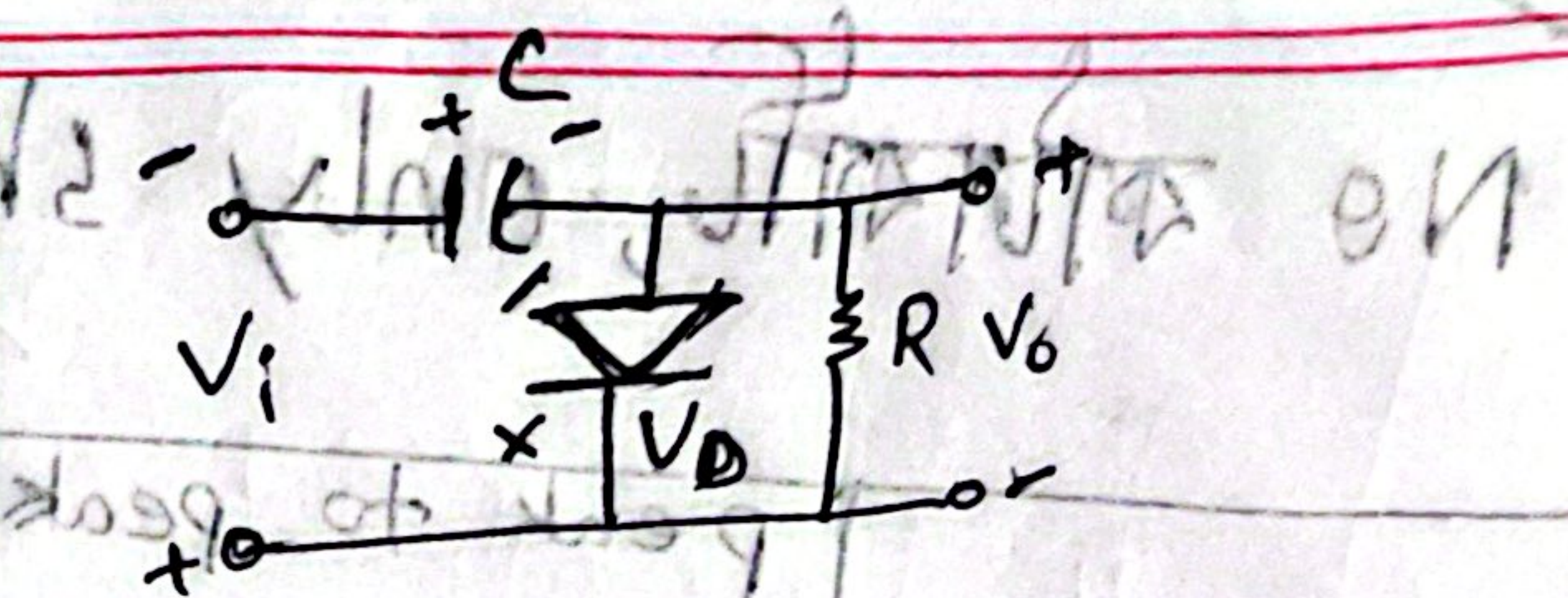
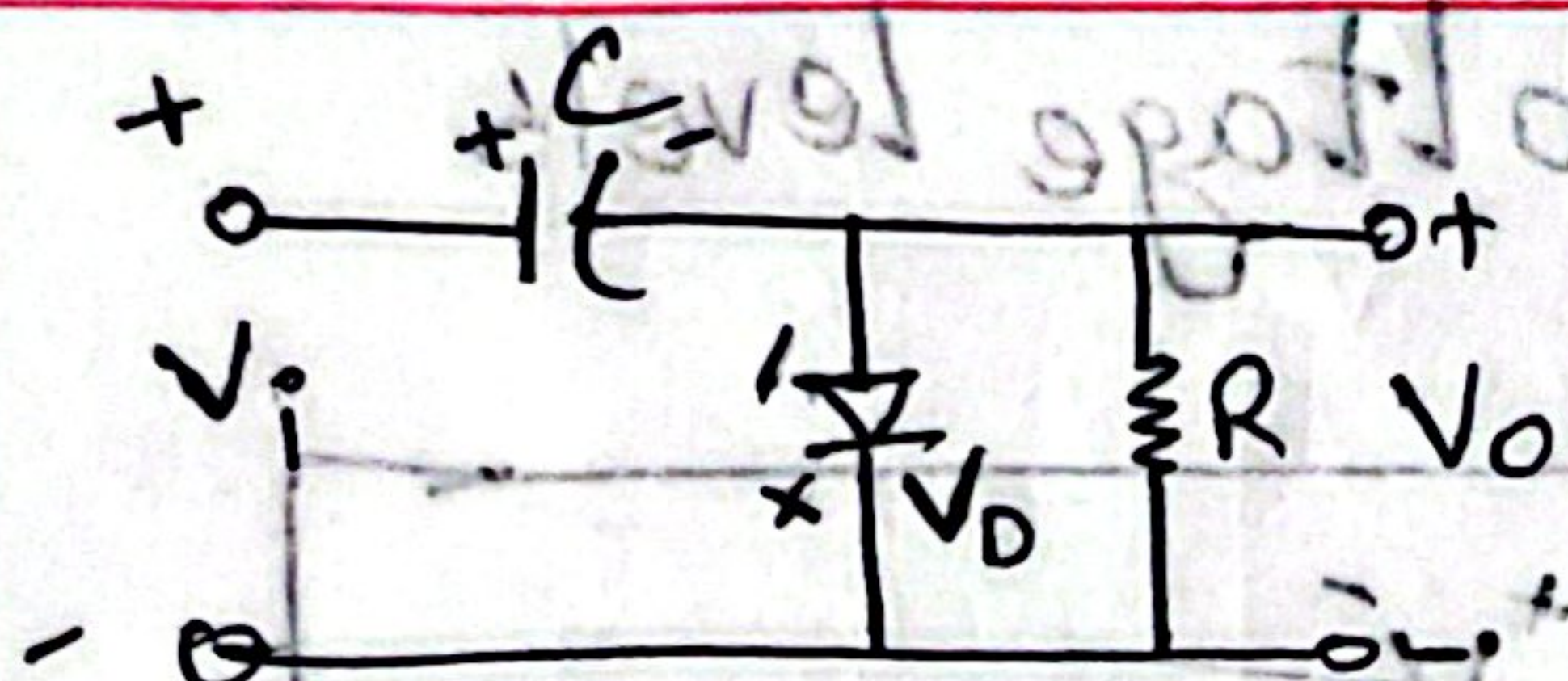
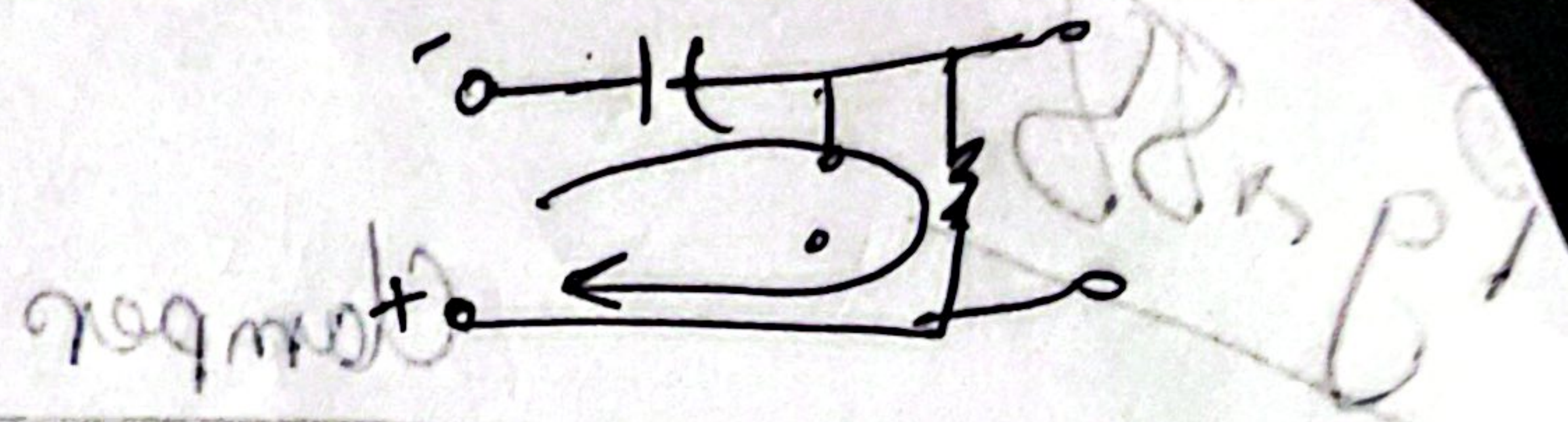
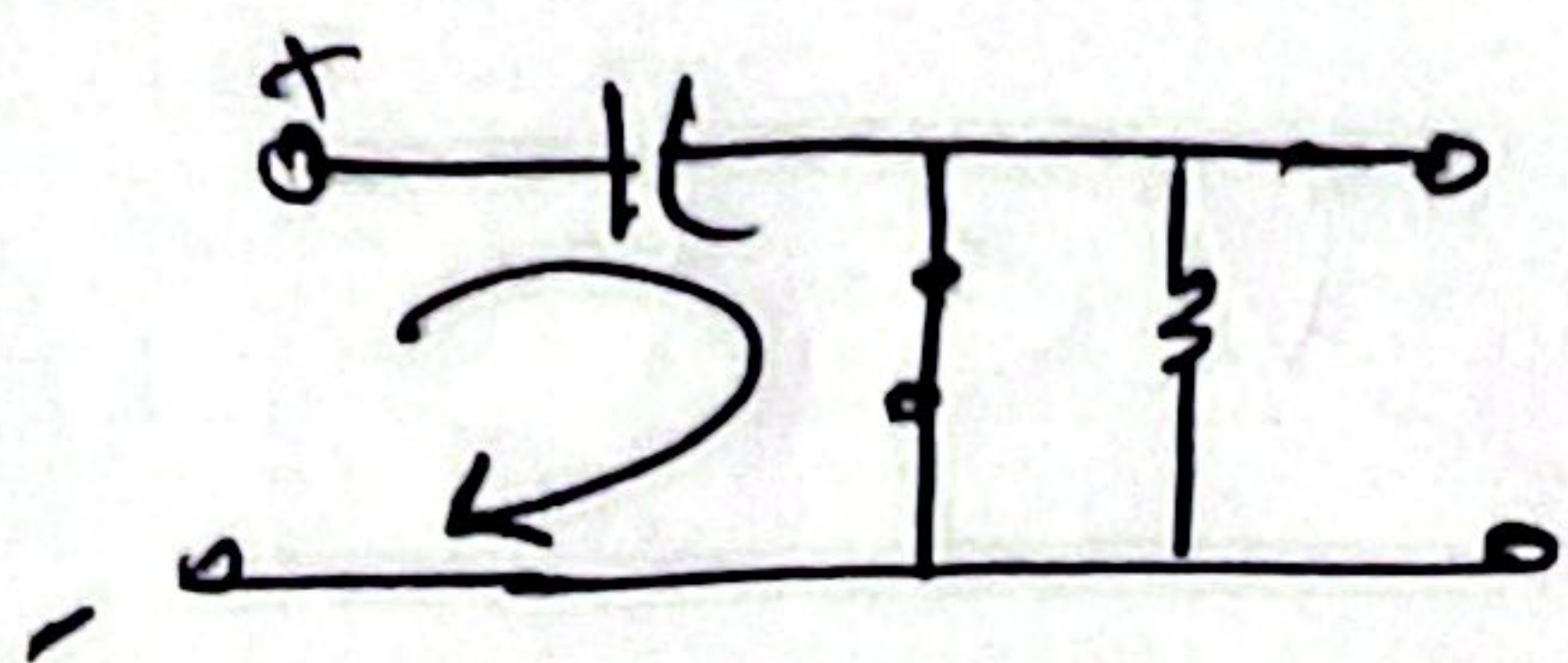
5

5

→ Use KVL in positive & negative half cycle.

→ Find Capacitor Voltage for First KVL, then V_o .

L , then V_o .



For positive half cycle:
KVL:
 $-V_i + C \neq 0$

$$\Rightarrow V_i = C$$

$$V_o = 0V$$

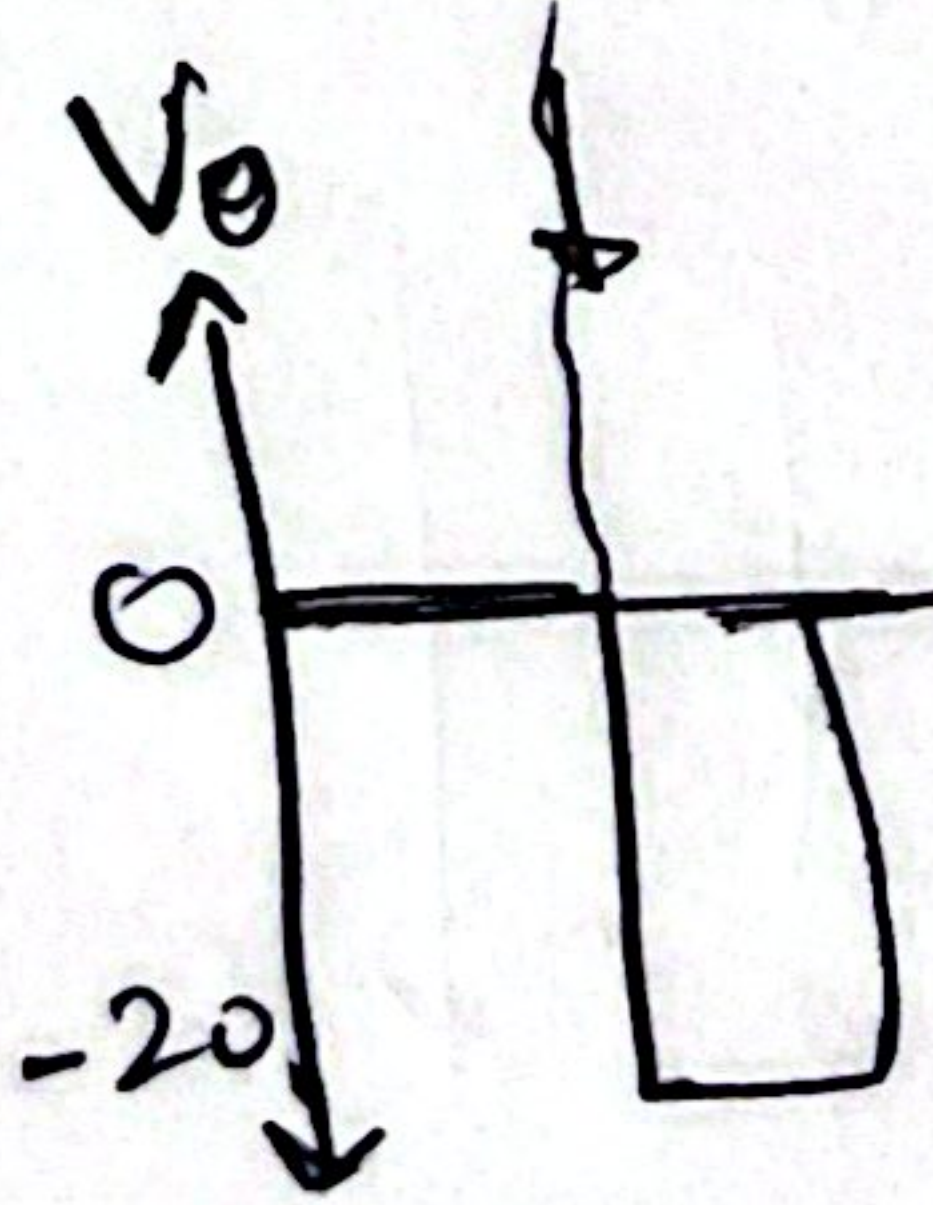
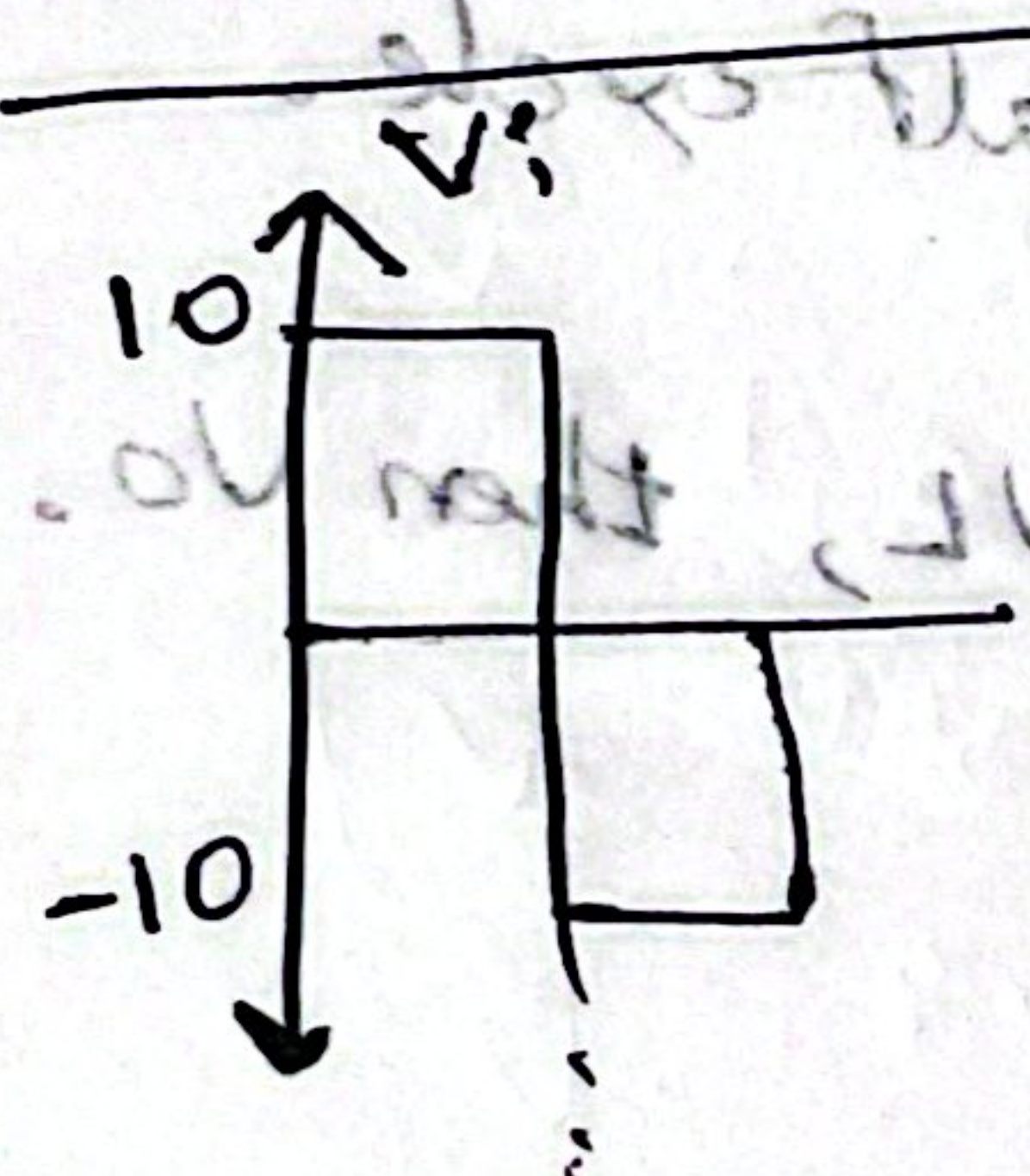
short circuit, all voltage in C.

For negative half cycle:
KVL:
 $+V_i + C + V_o = 0$

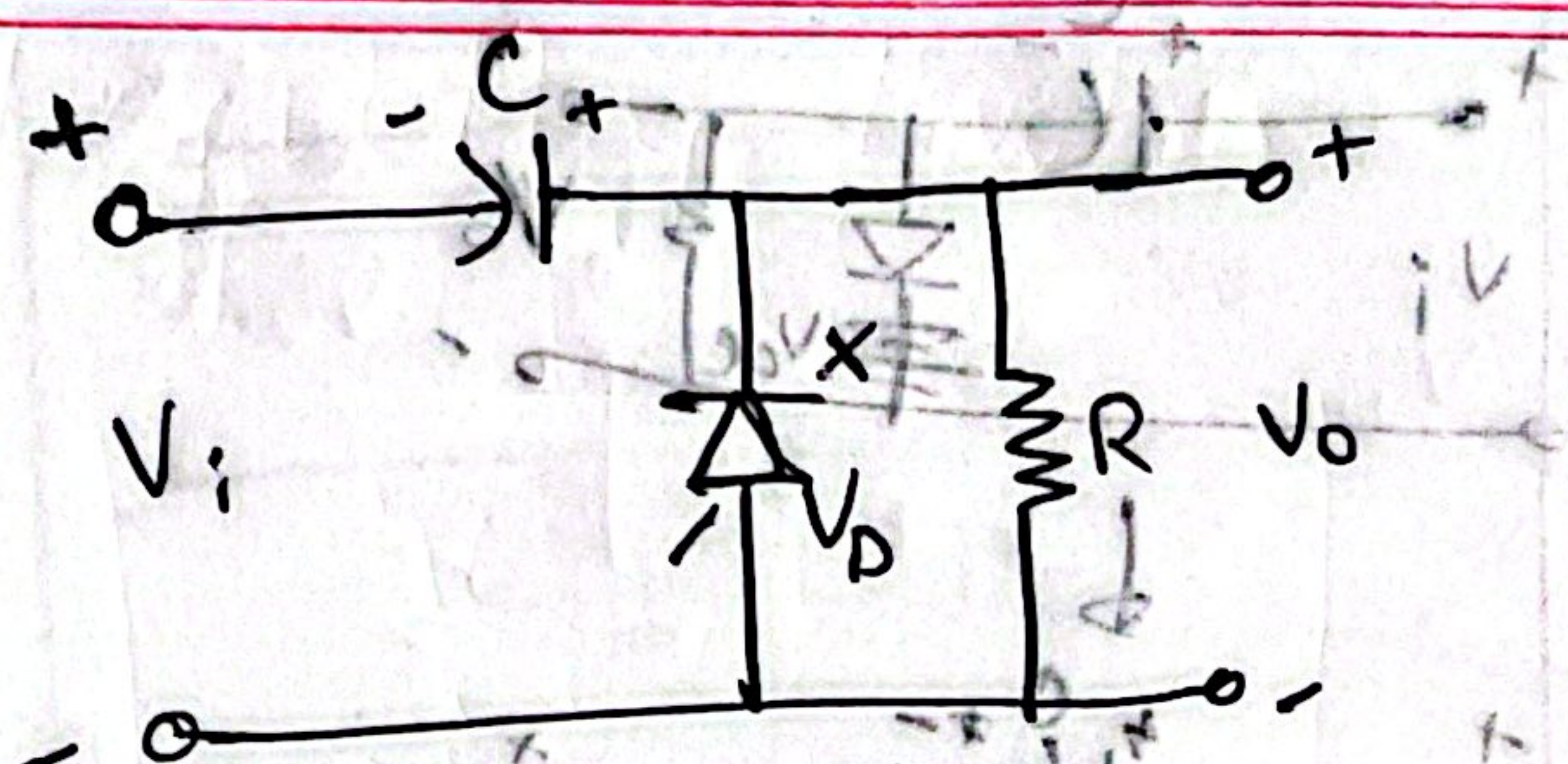
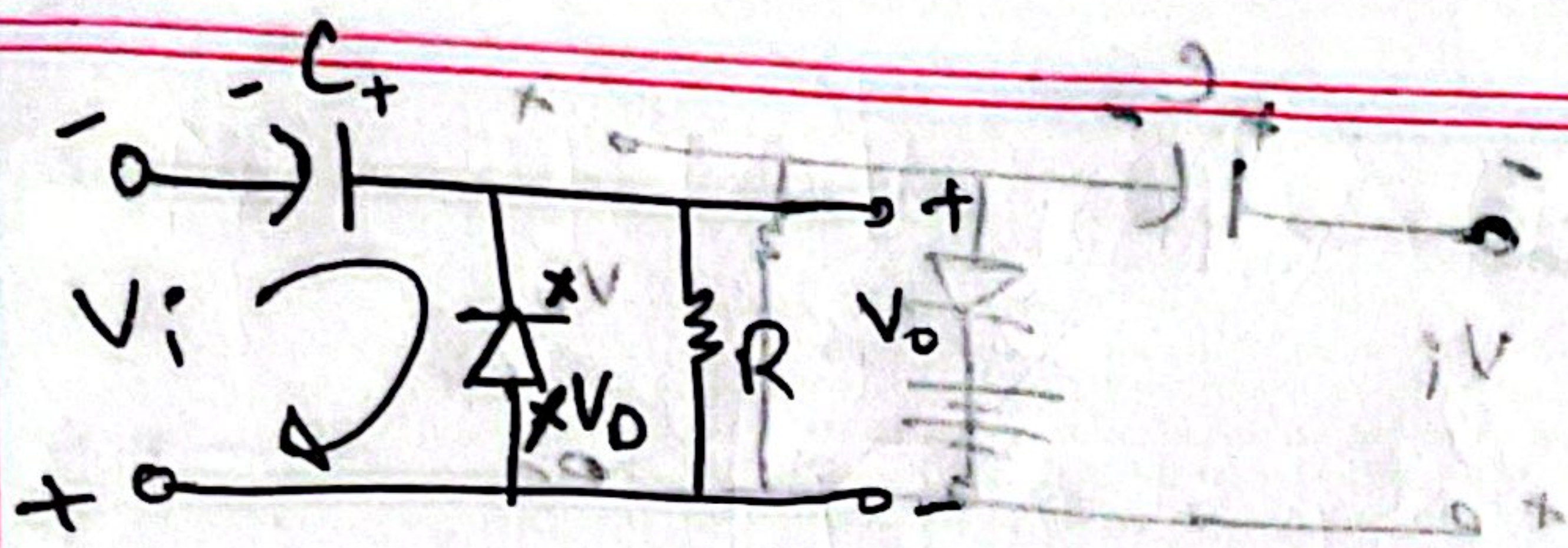
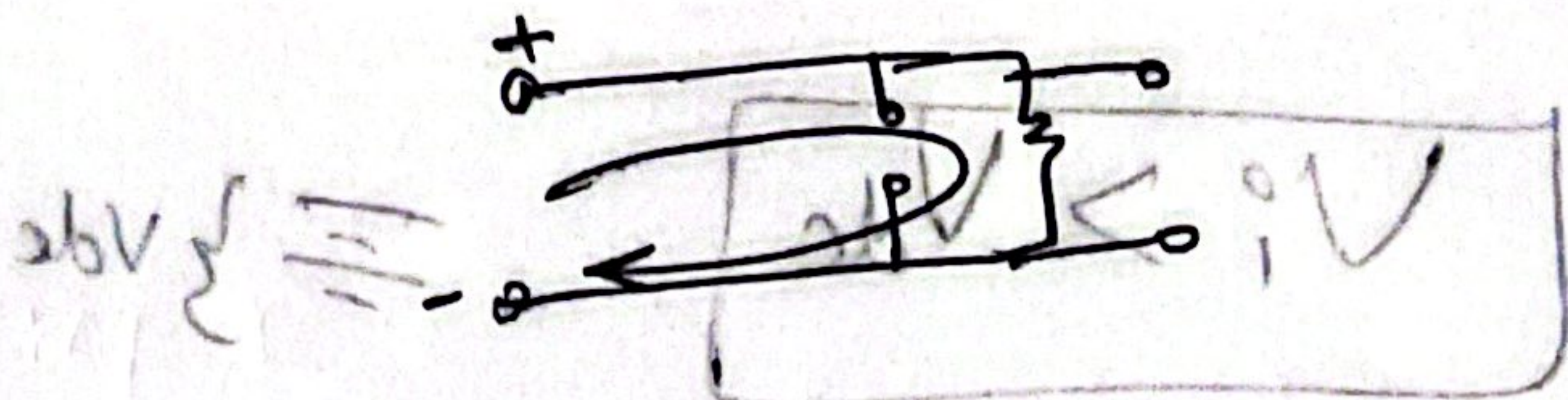
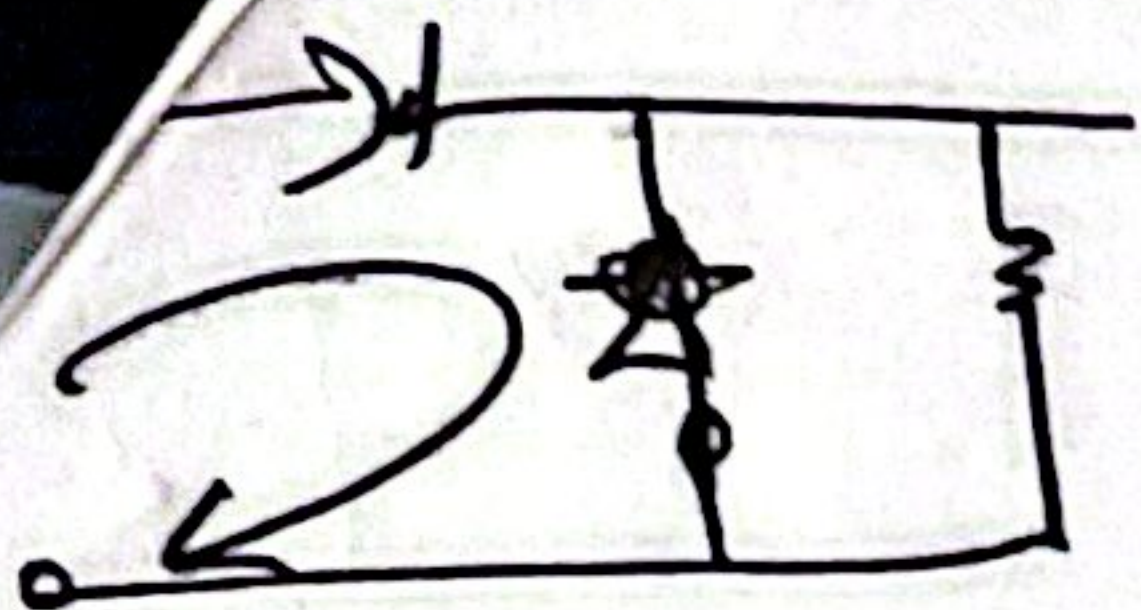
$$\Rightarrow V_o = -V_i - C$$

$$V_o = -V_i - V_i$$

$$V_o = -2V_i$$



Find capacitor voltage for first KVL.
Then V_o for 2nd KVL.
Use KVL in positive & negative half cycle.



For negative half cycle

For positive half cycle

KVL in short loop:

KVL in open loop:

$$+V_i - C = 0$$

$$-V_i - C + V_o = 0$$

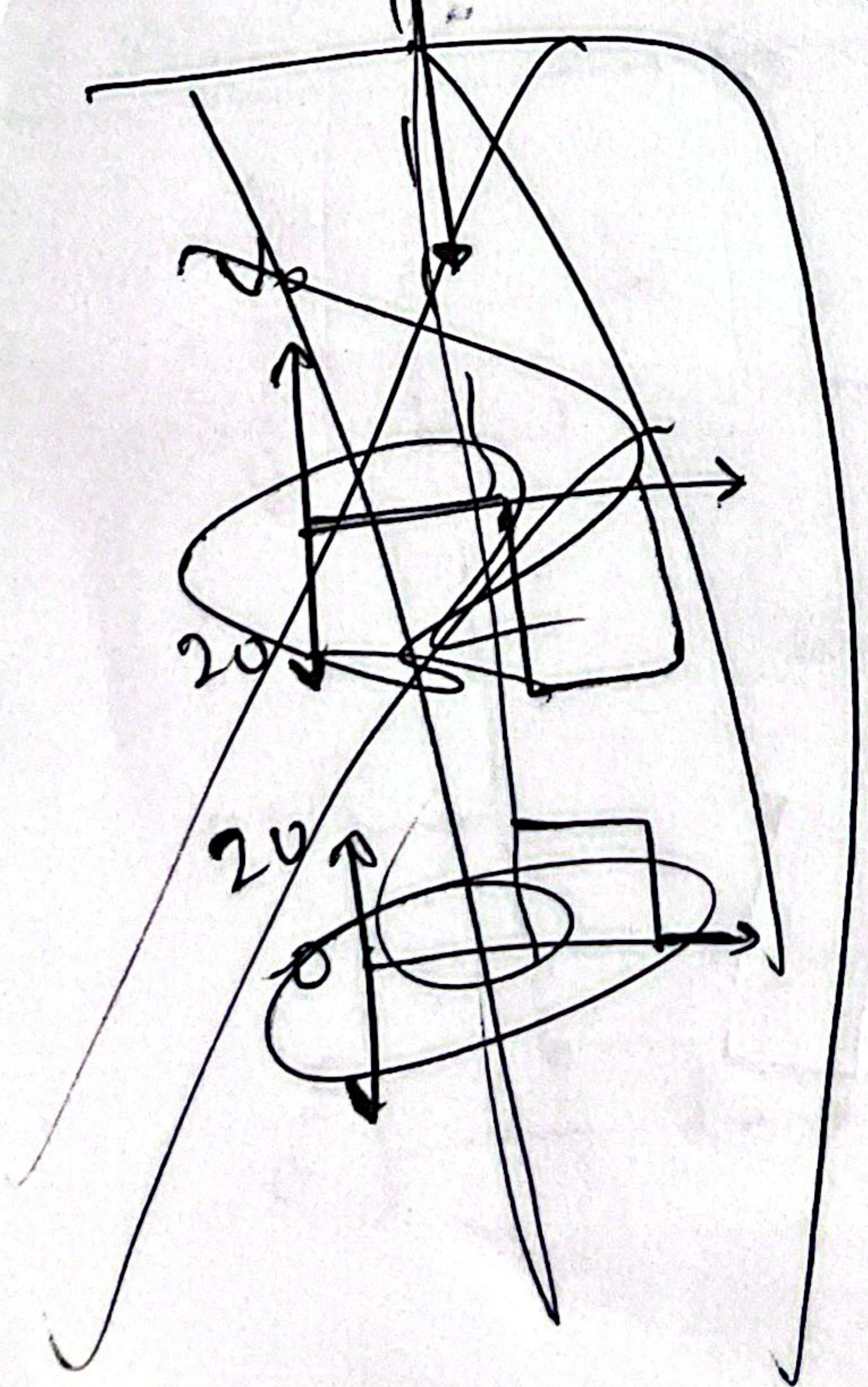
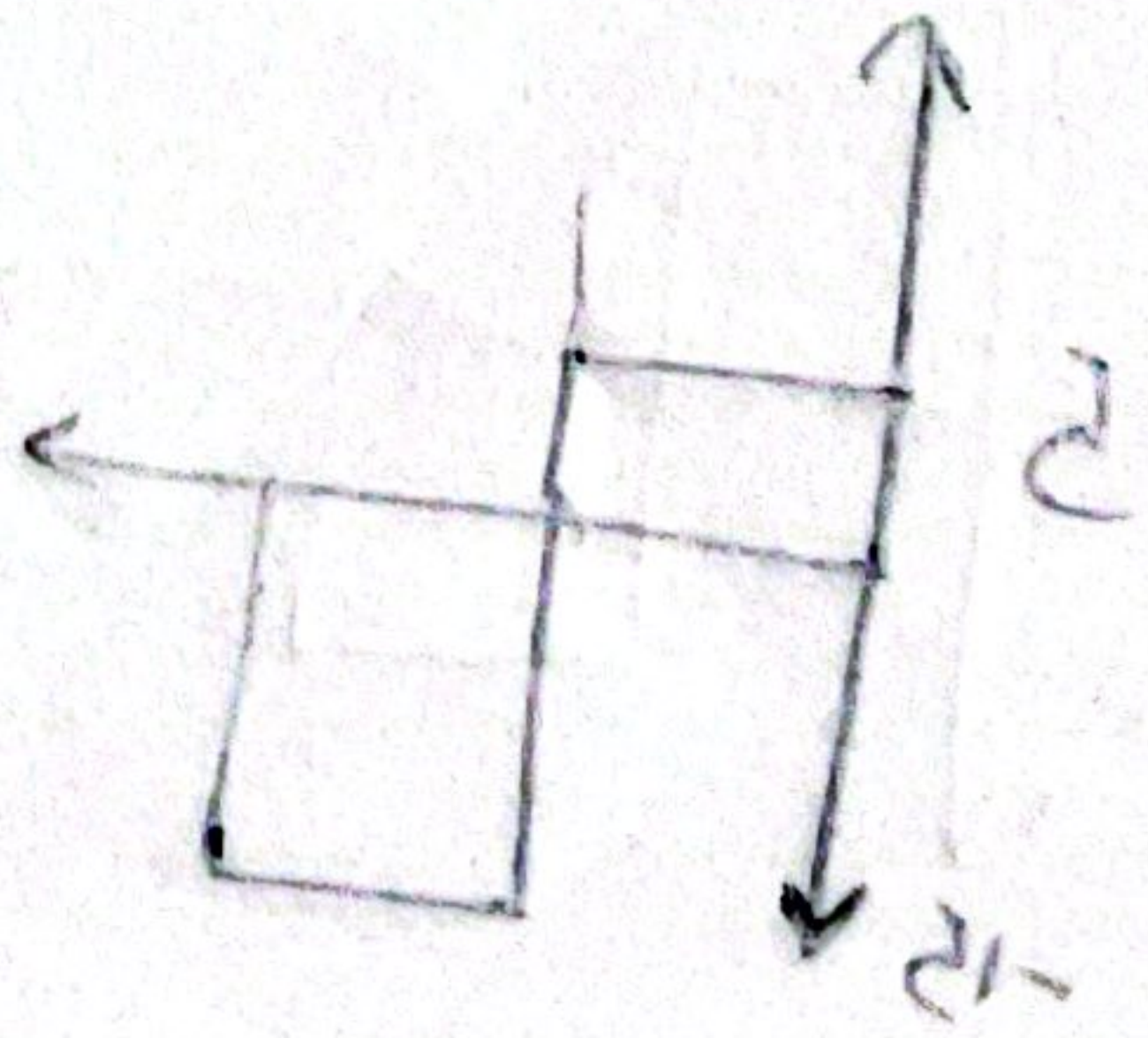
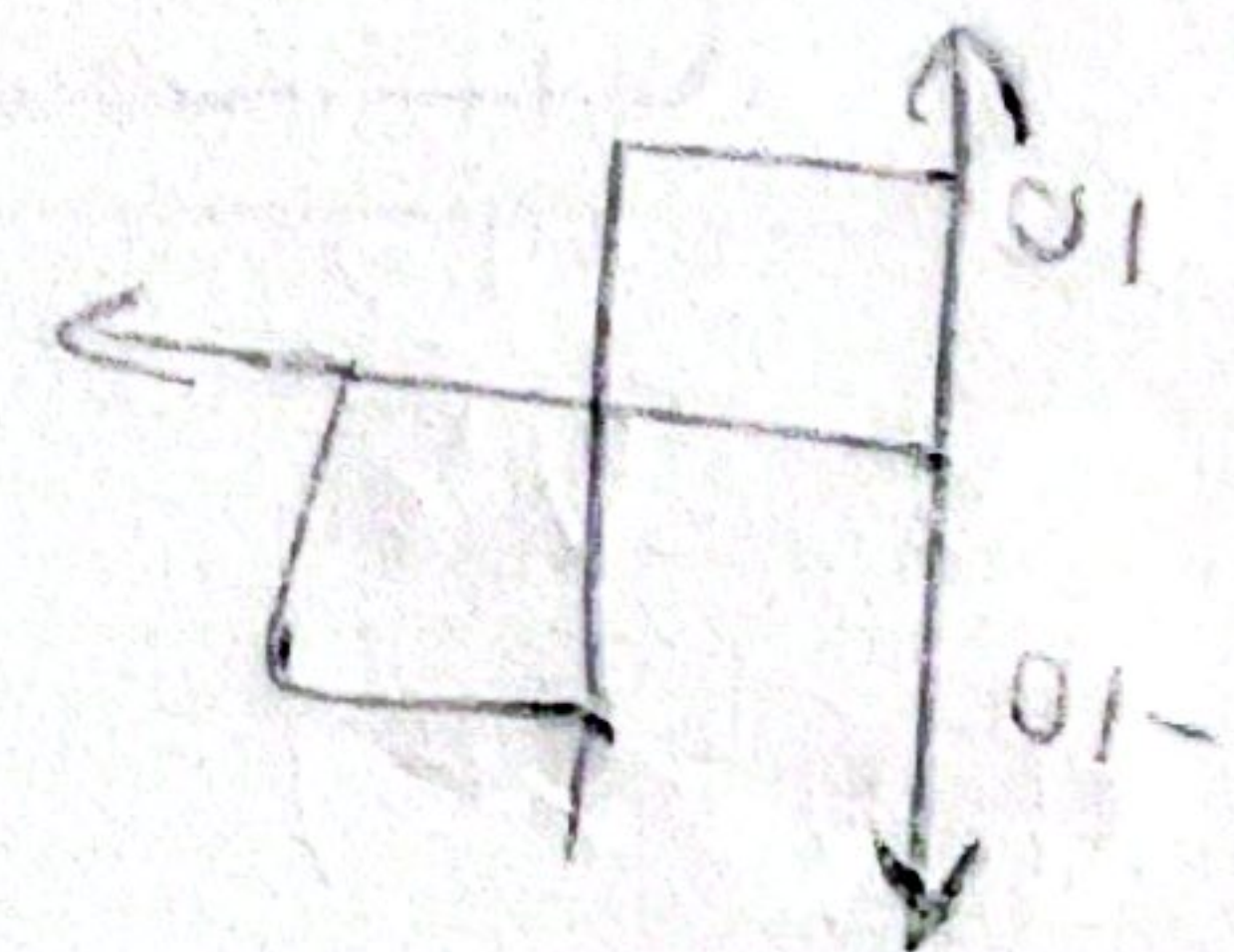
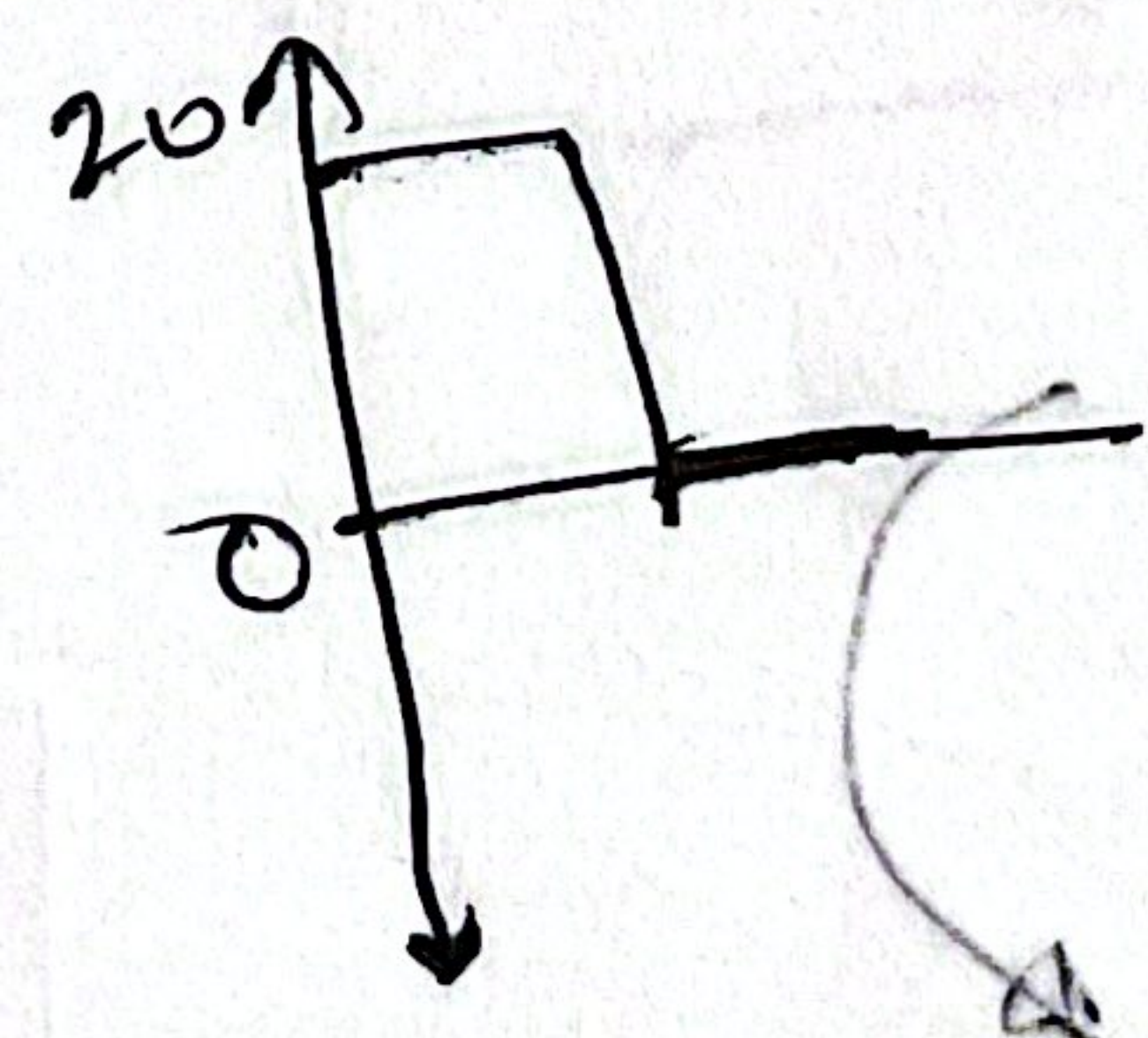
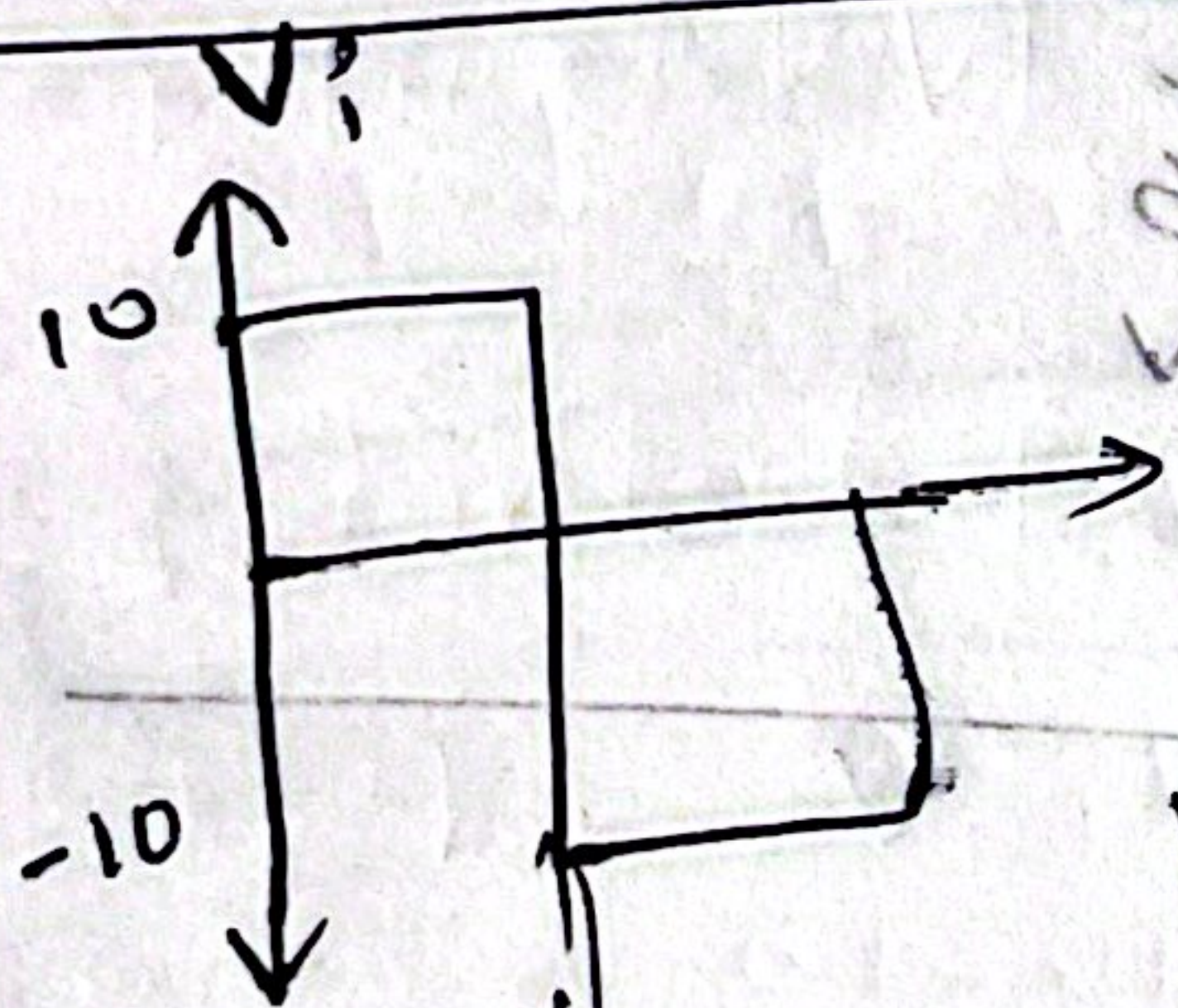
$$\Rightarrow V_i = C$$

$$\Rightarrow V_o = C + V_i$$

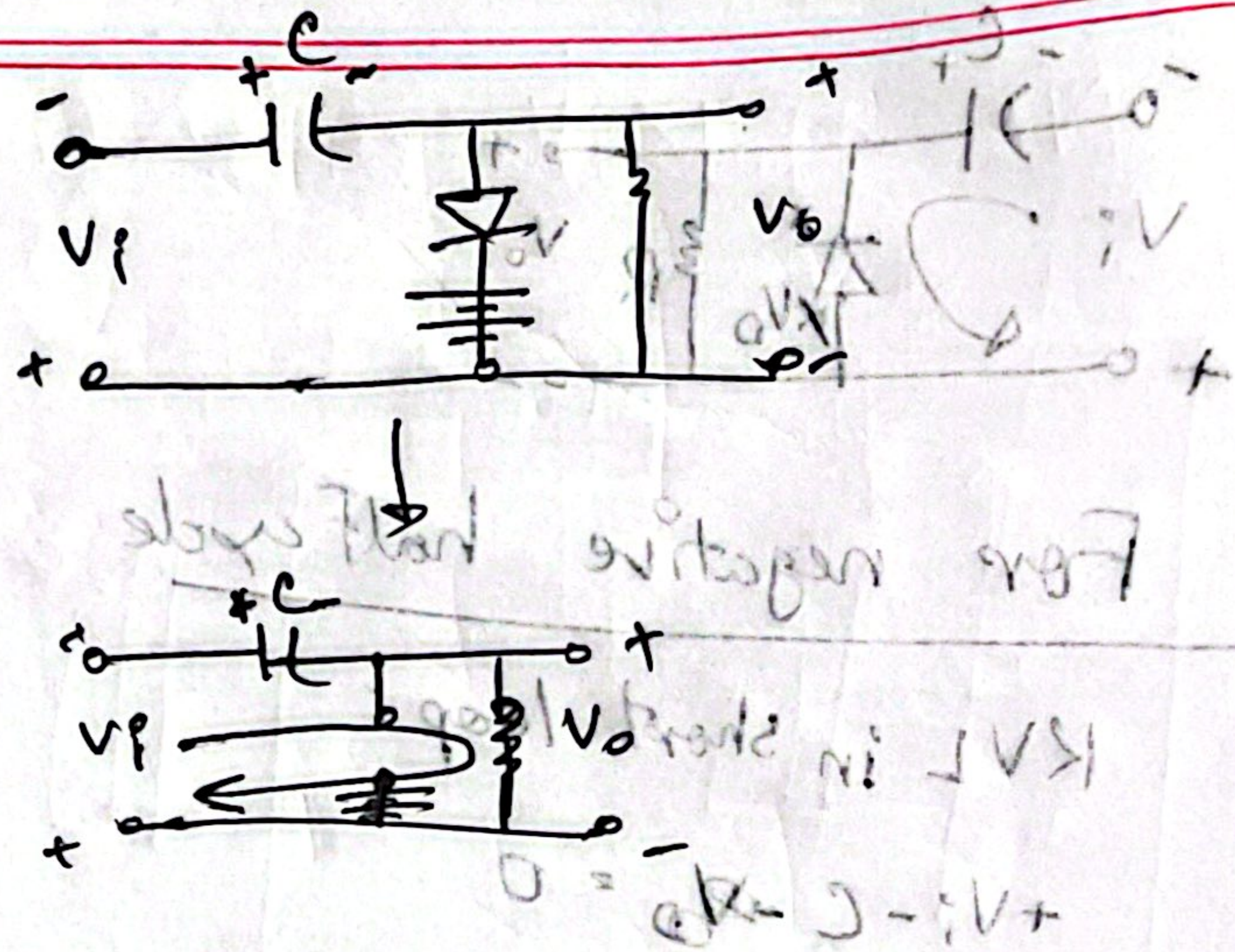
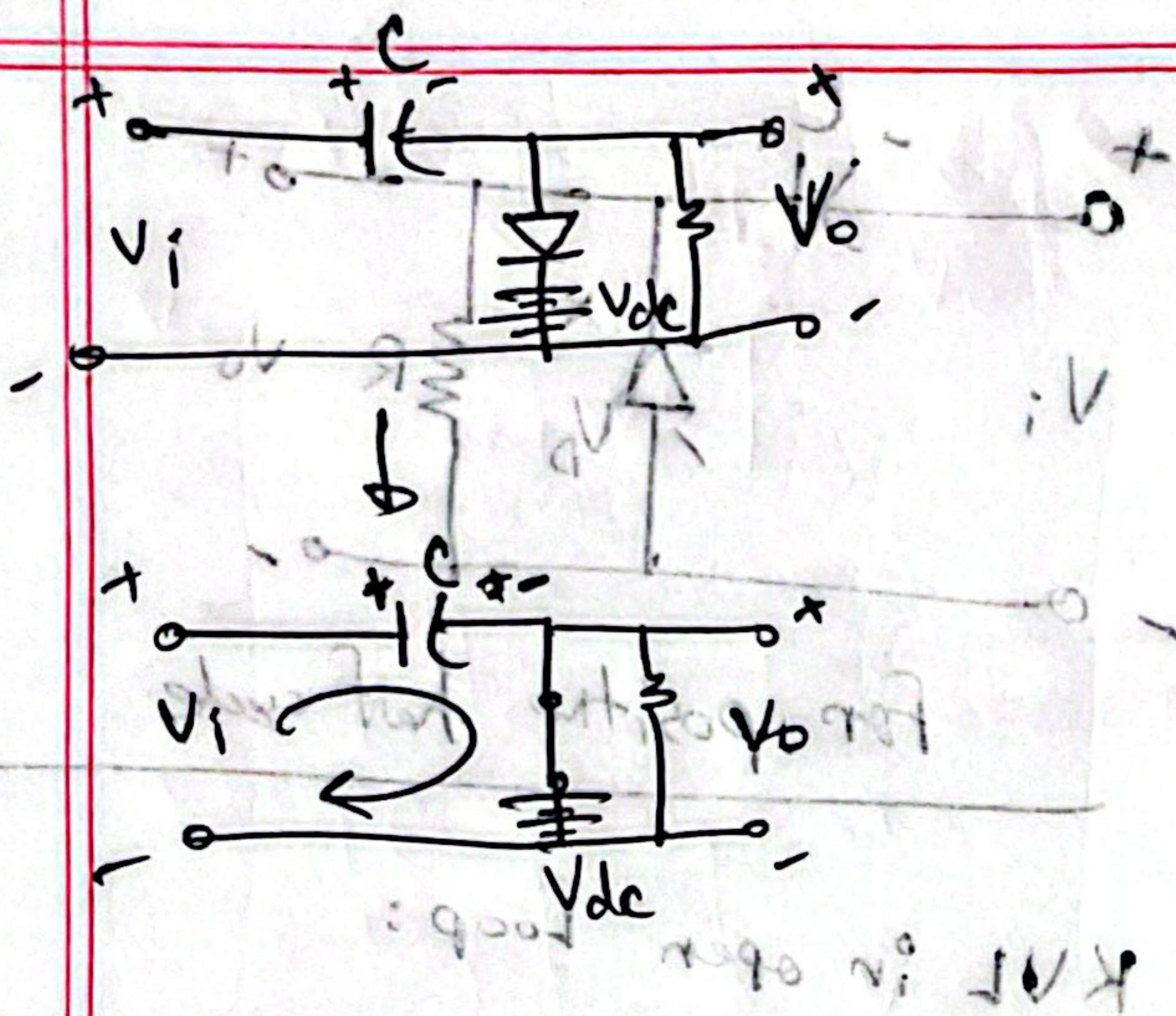
$$V_o = 0V$$

$$V_o = 0V$$

$$V_o = 0V$$



$$V_i > V_{dc} \Rightarrow \equiv V_{dc}$$



For positive half cycle
KVL in short loop

For negative half cycle
KVL in open loop

$$-V_i + C + V_{dc} = 0$$

$$\Rightarrow V_i + C + V_{dc} = 0$$

$$\Rightarrow C = -V_i - V_{dc}$$

$$= -10 - 5$$

$$= -15$$

$$+V_i + C + V_o = 0$$

$$\Rightarrow V_o = -V_i - C$$

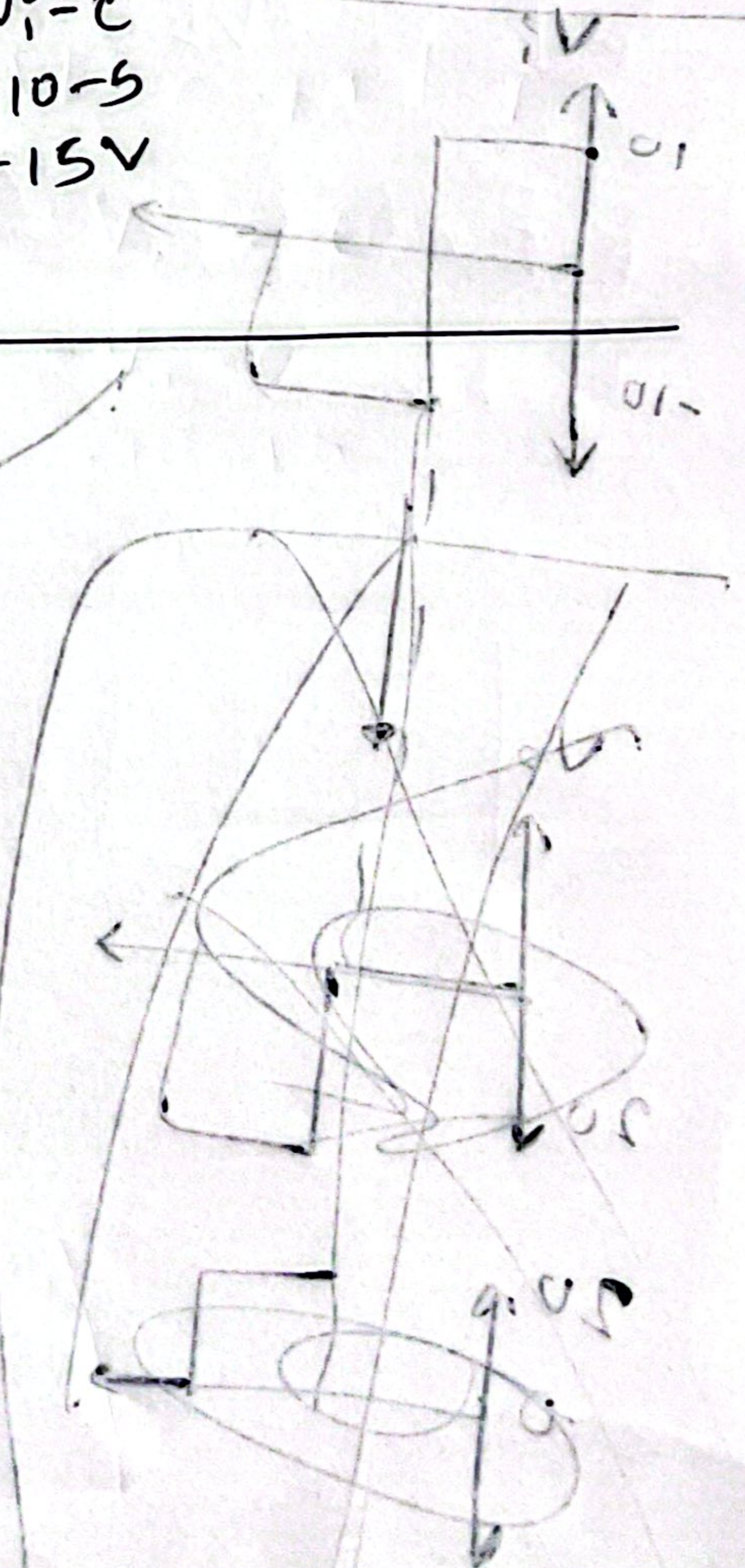
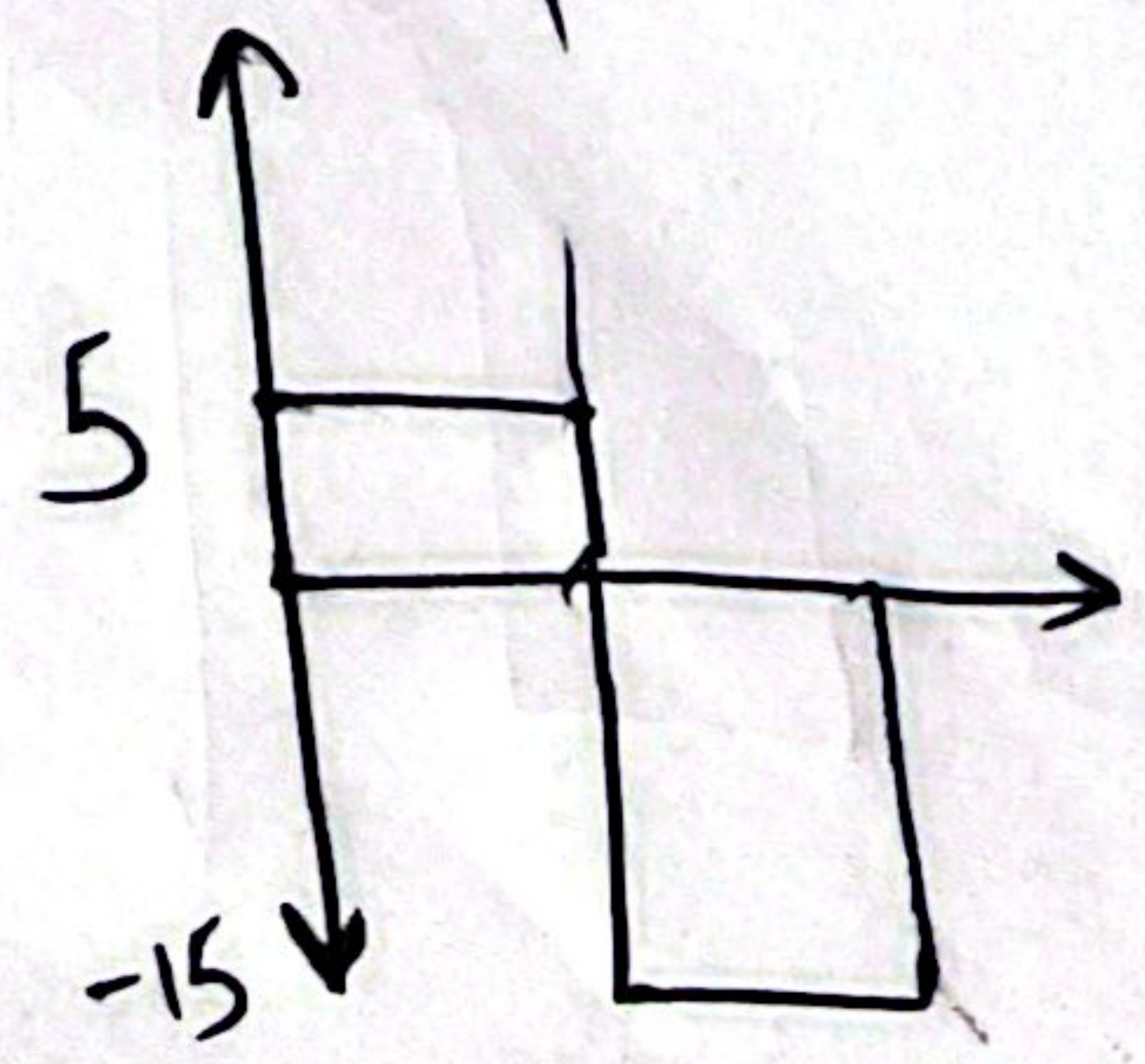
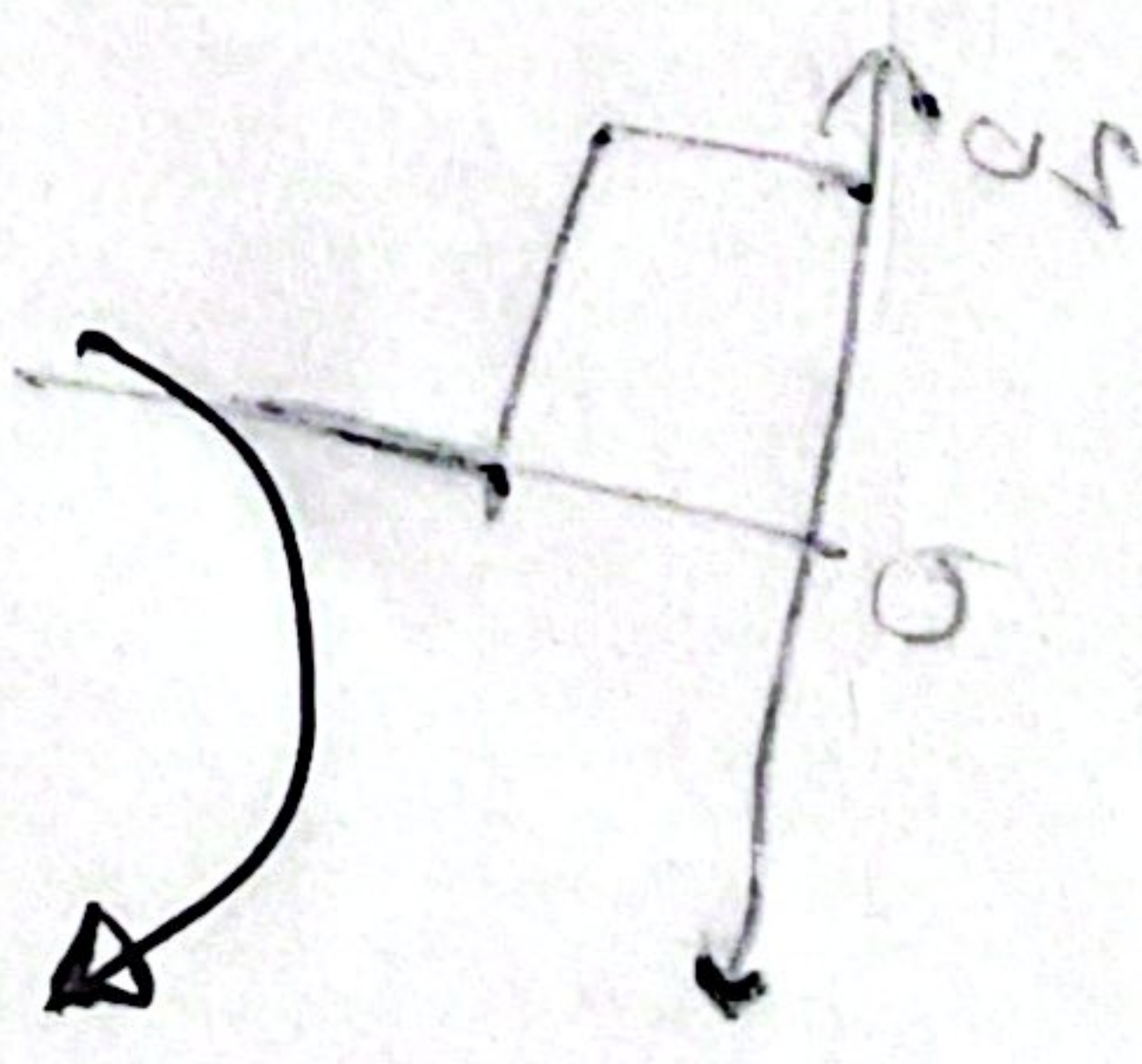
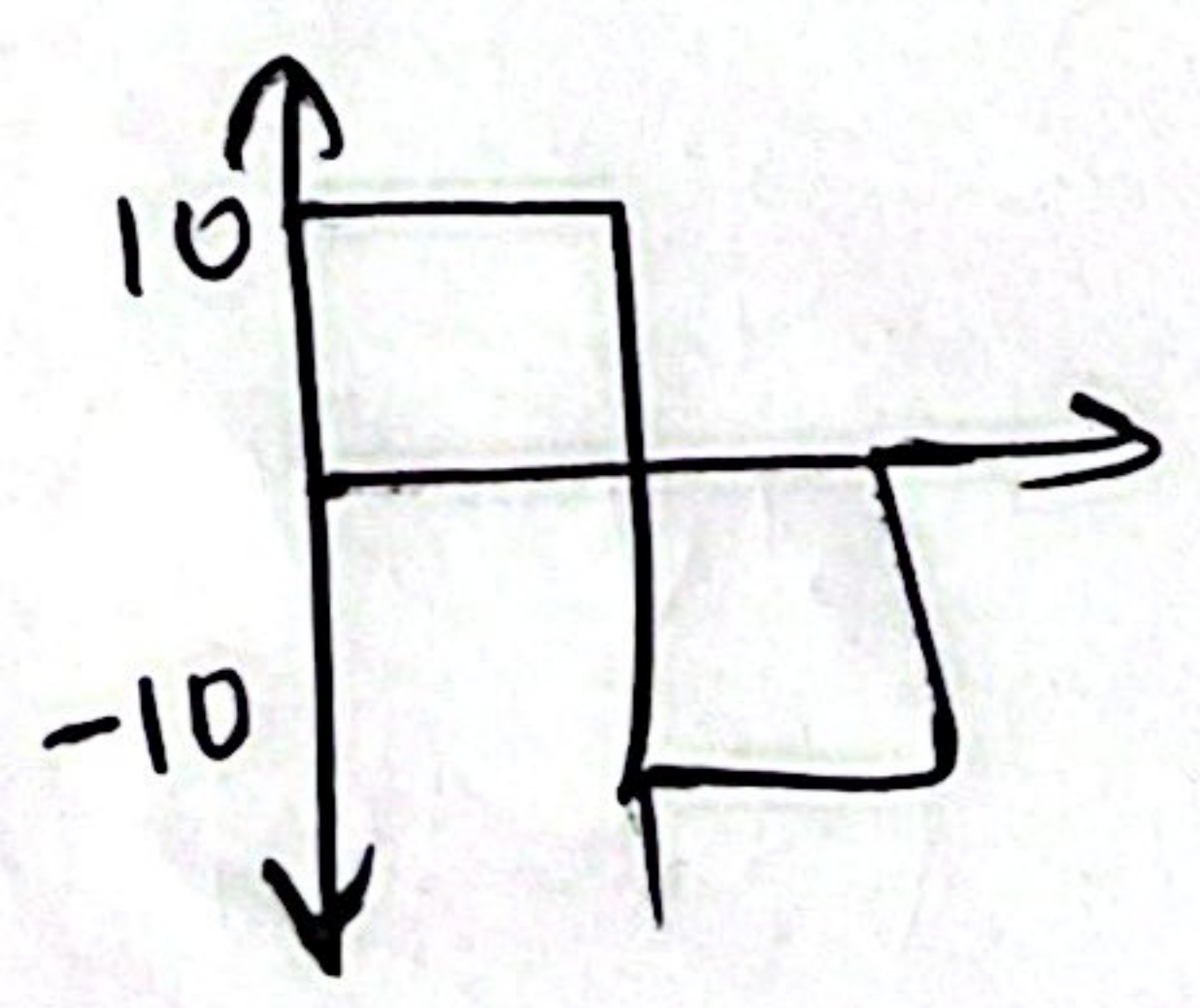
$$= -10 - (-15)$$

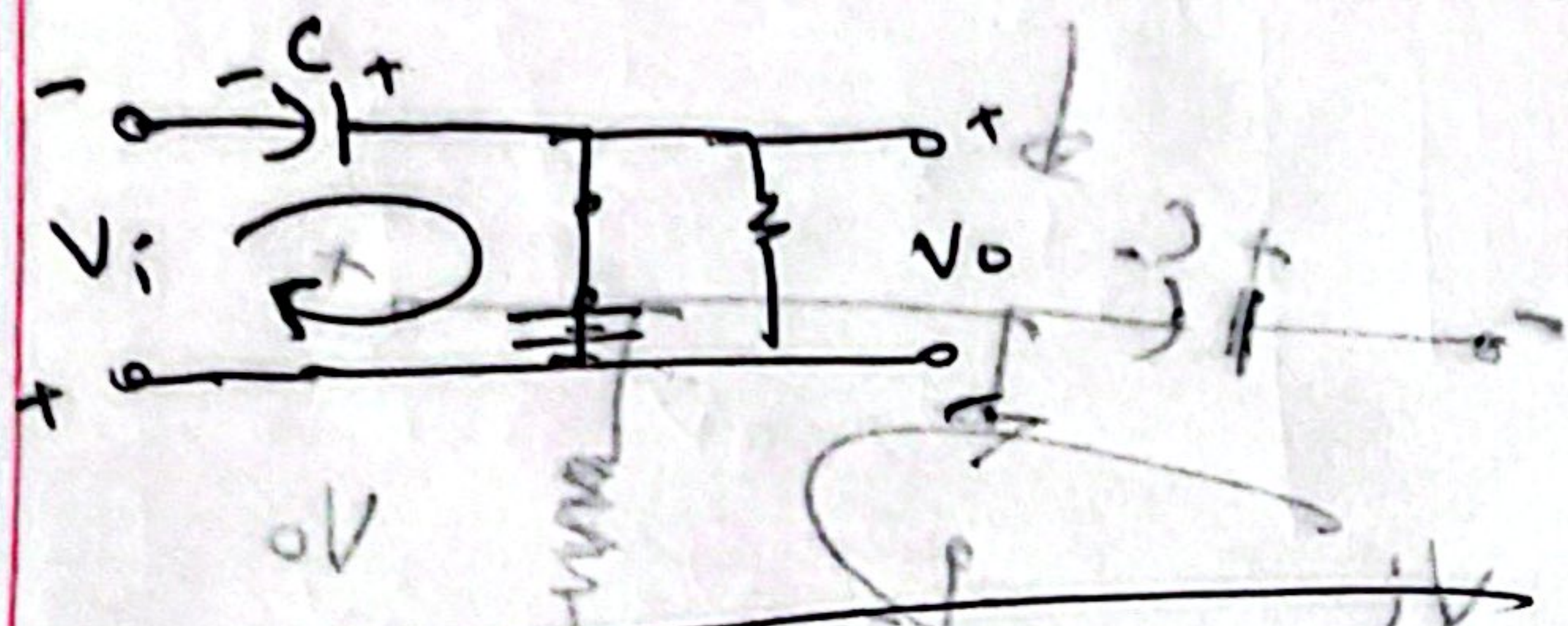
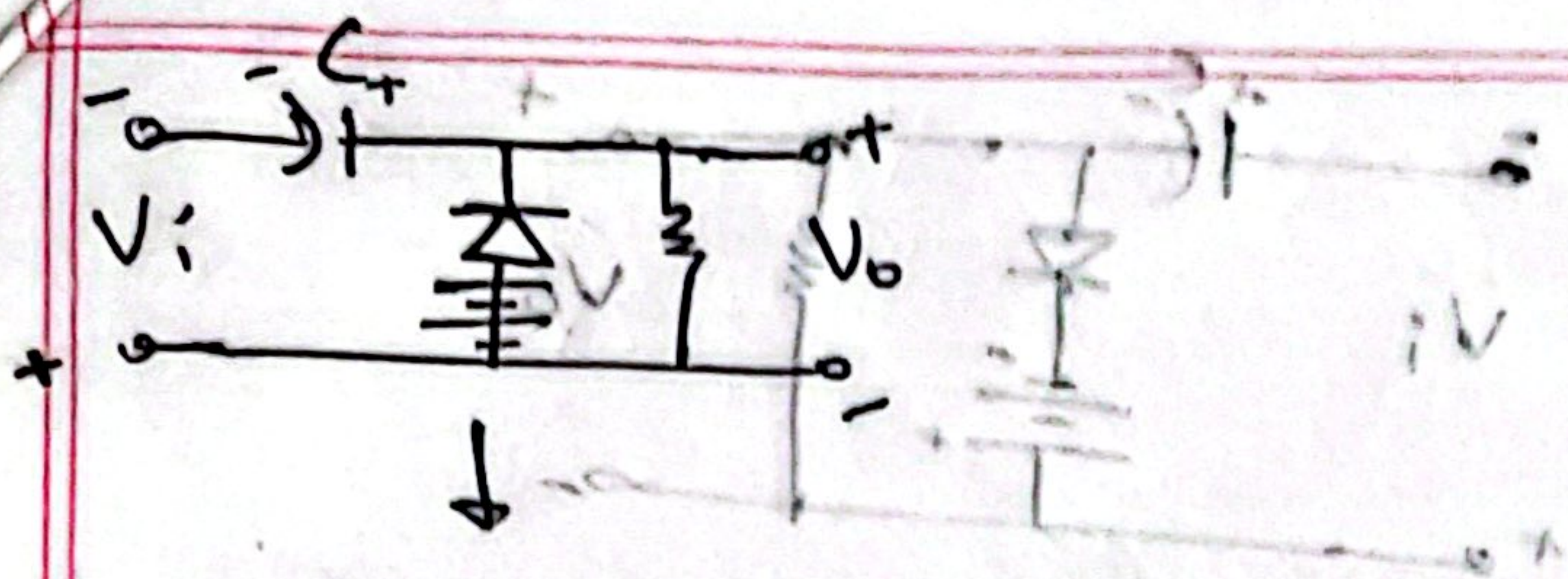
$$= -10 + 15$$

$$= 5$$

$$V_o = V_{dc}$$

$$V_o = 5V$$





For negative half cycle

KVL in short loop

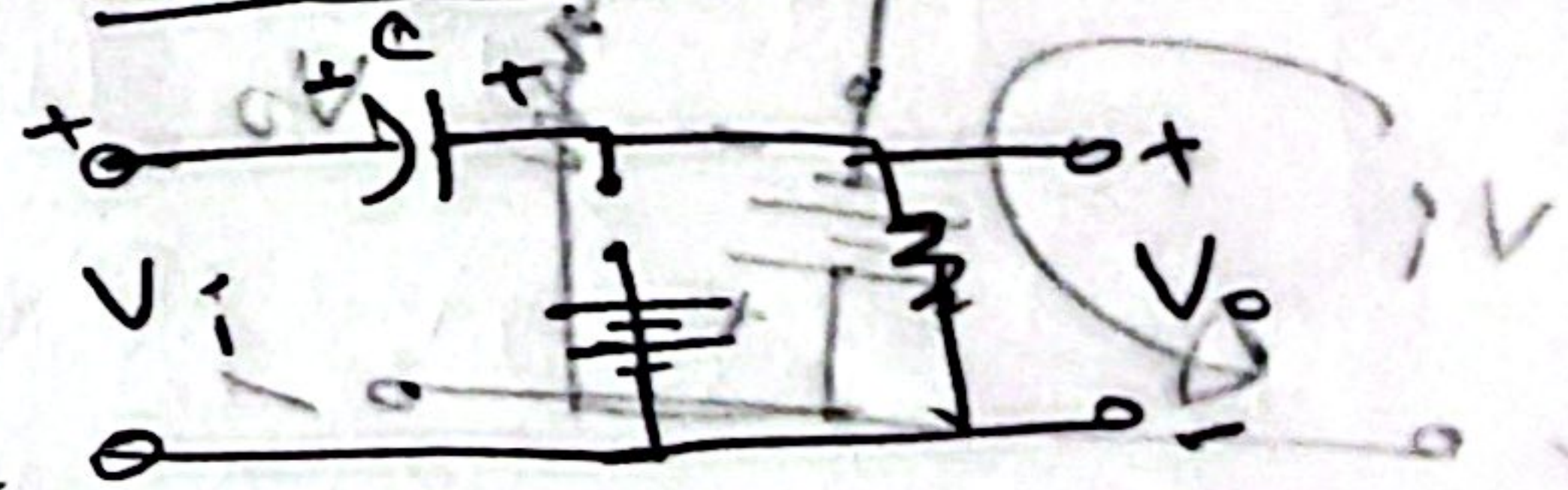
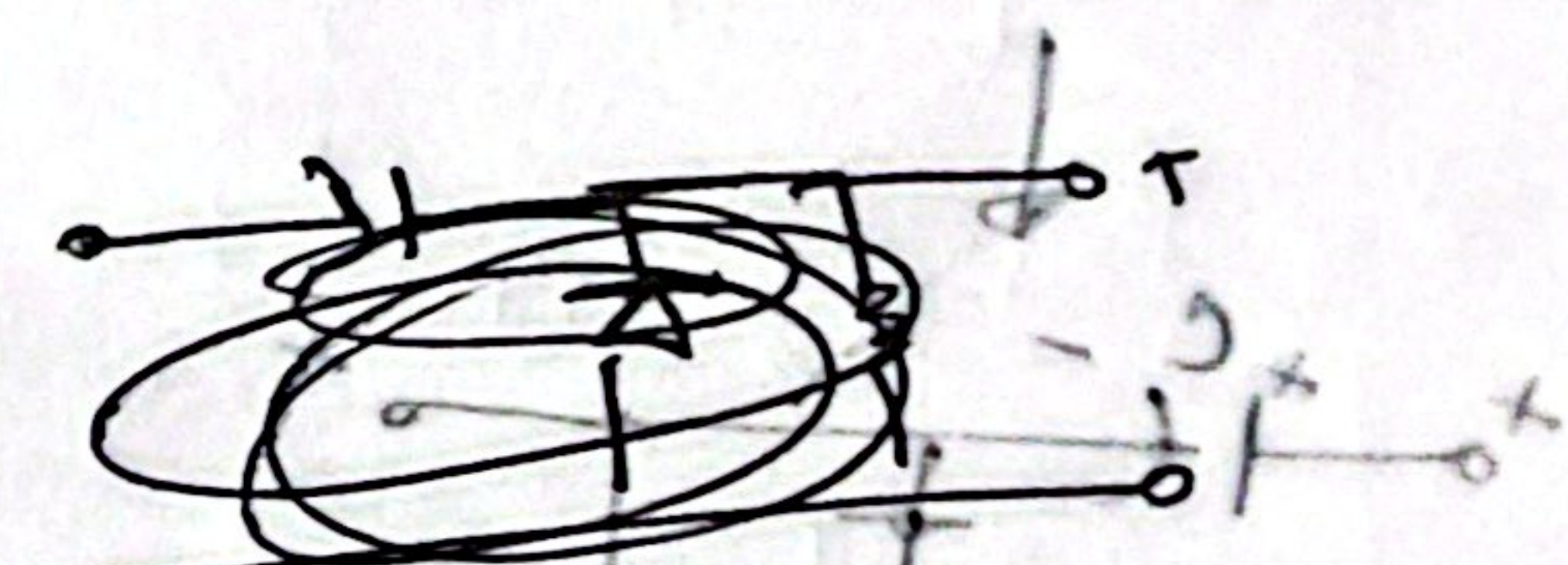
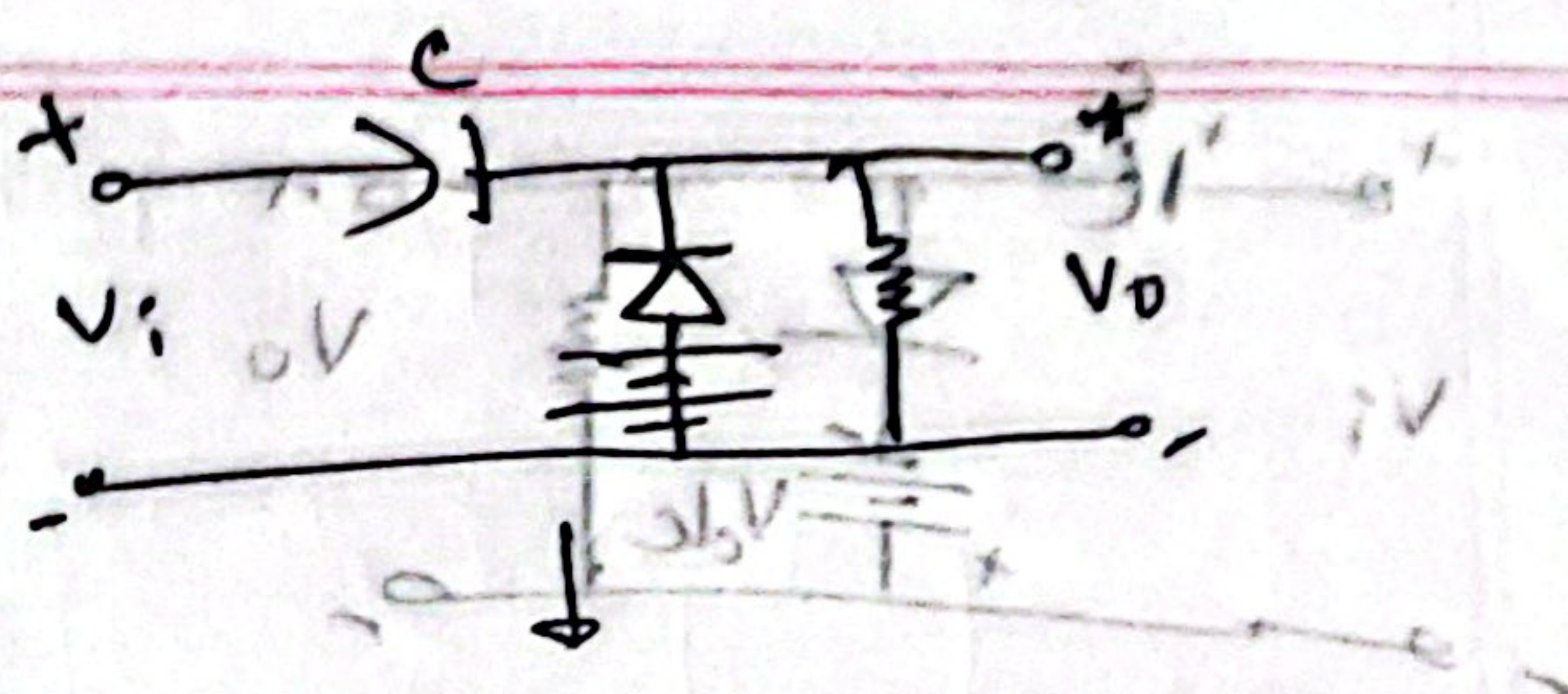
$$+V_i - C + V_{dc} = 0$$

$$C = V_i + V_{dc}$$

$$C = 10 + 5 = 15V$$

$$V_o = V_{dc}$$

$$V_o = 5V$$



For positive half cycle

KVL in open loop

$$-V_i - C + V_o = 0$$

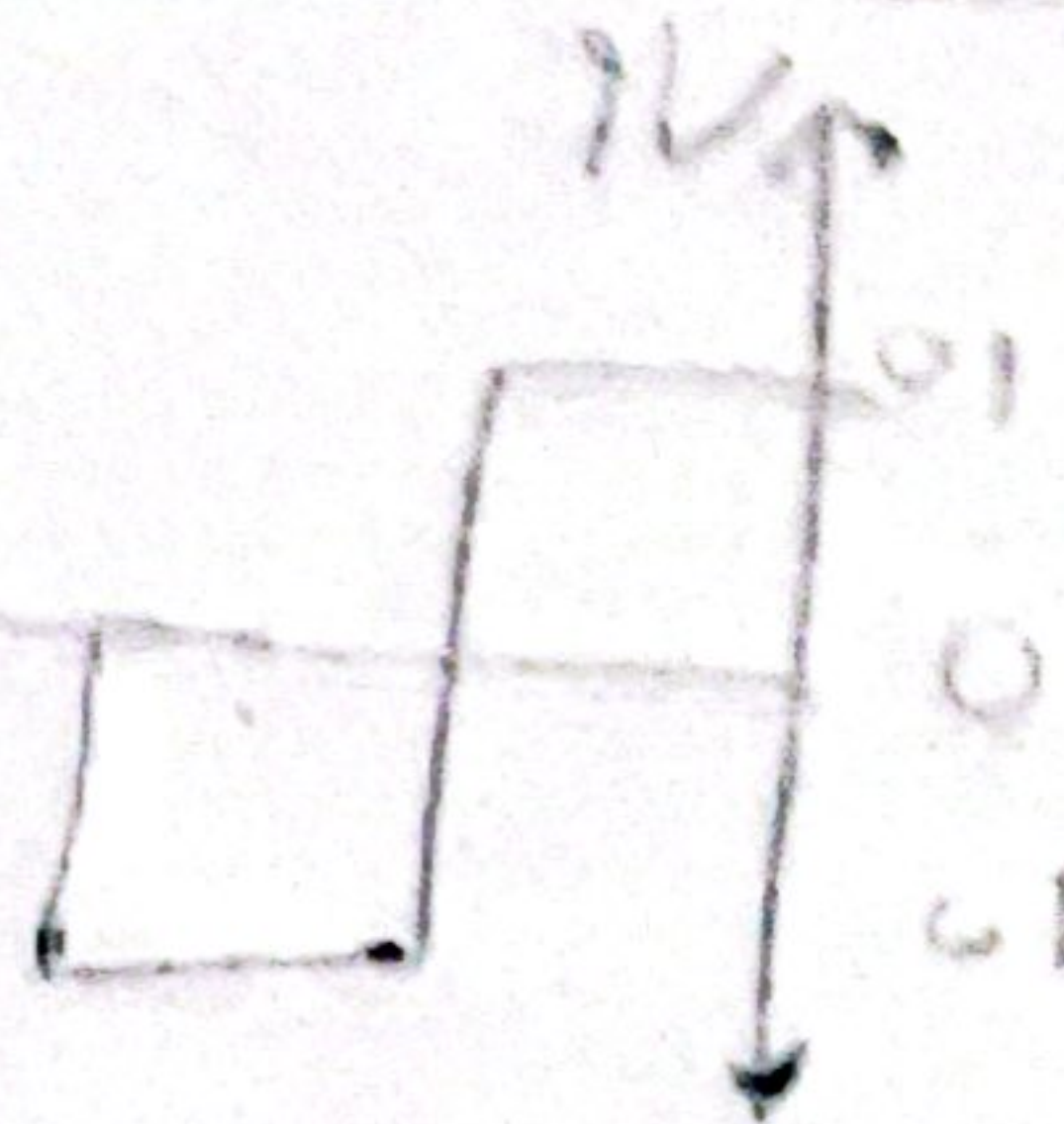
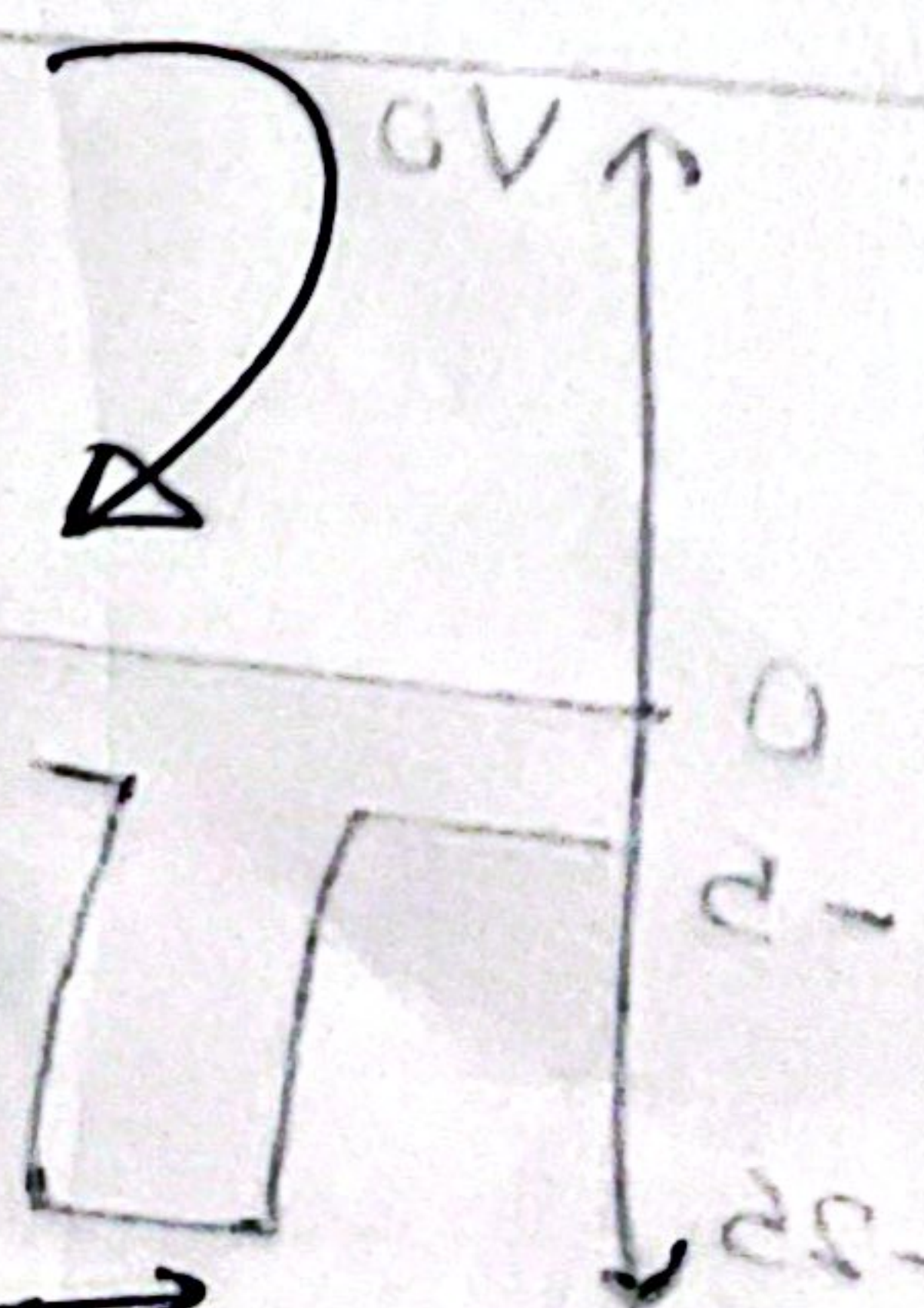
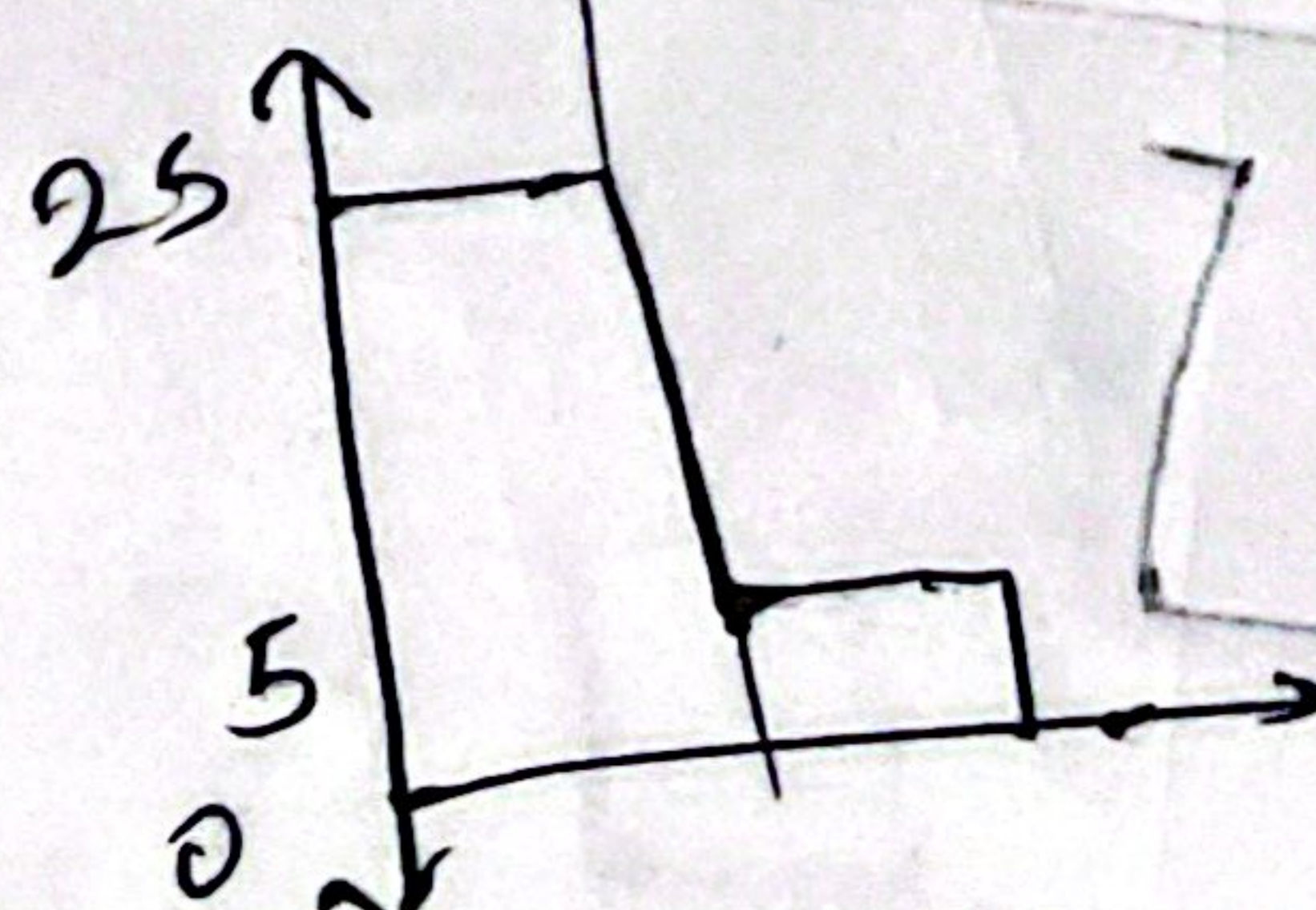
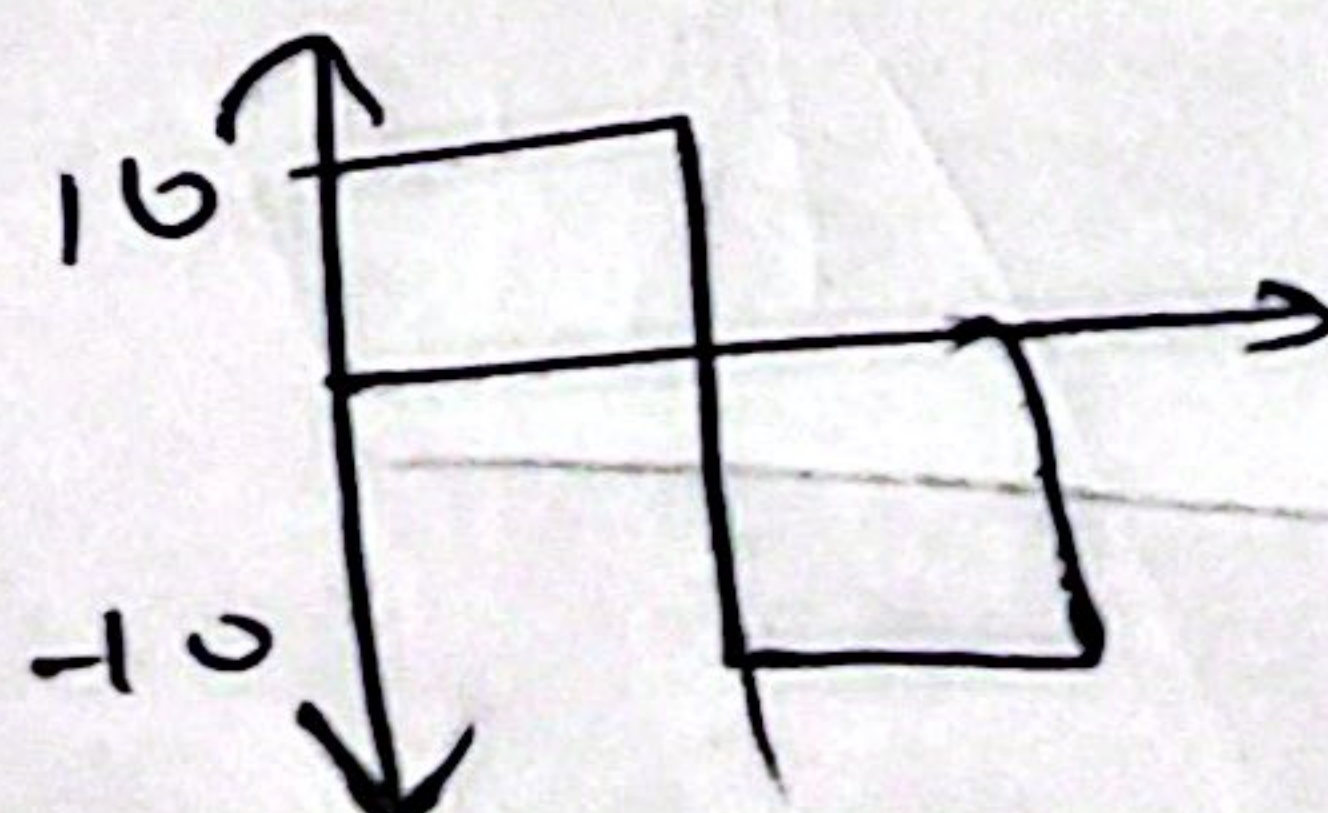
$$V_o = V_i + C$$

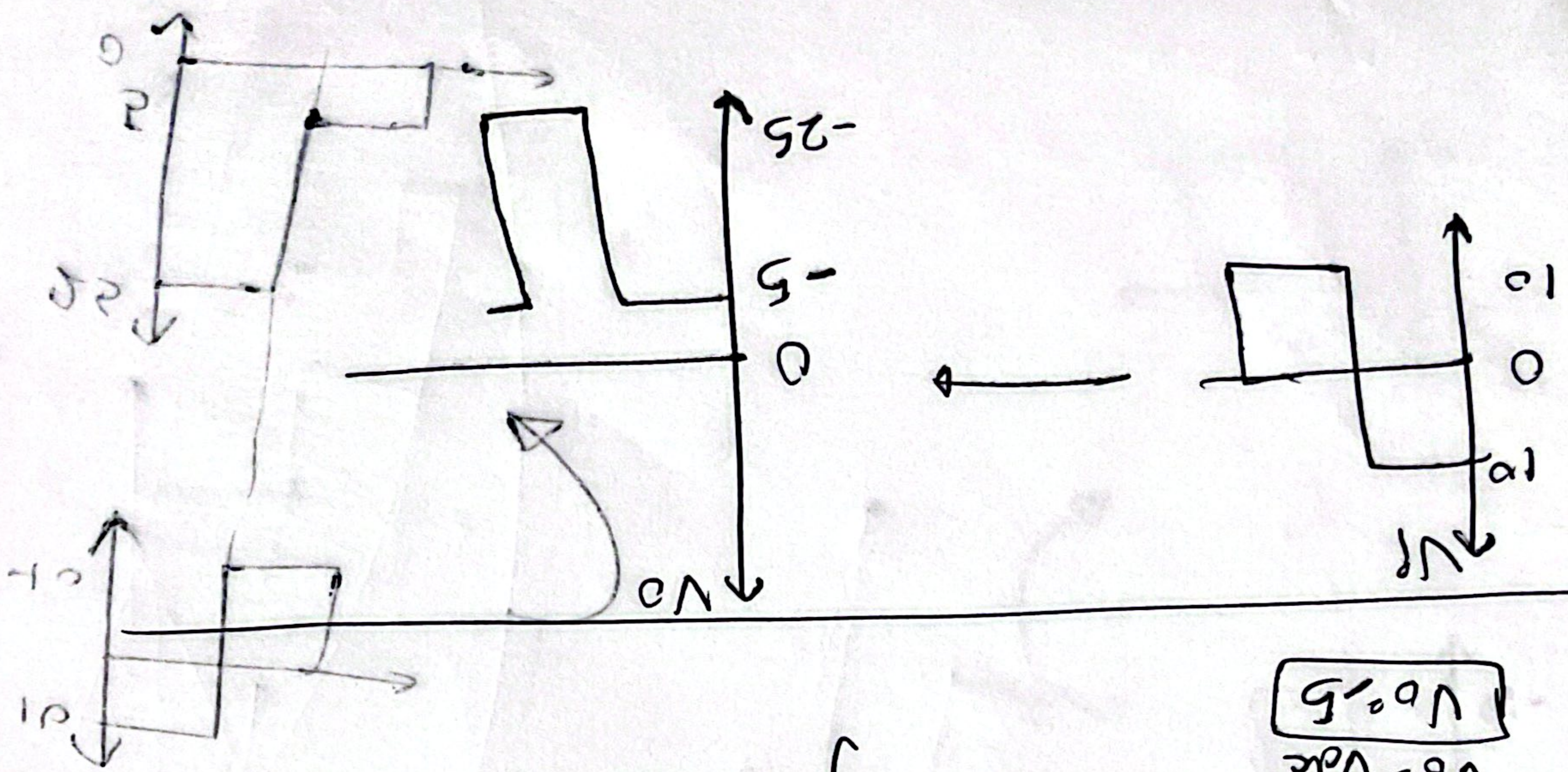
$$= 10 + 15$$

$$V_o = 25V$$

$$V_o = 25V$$

$$V_o = 25V$$





$$V_o = 5V$$

$$V_o = V_{dc}$$

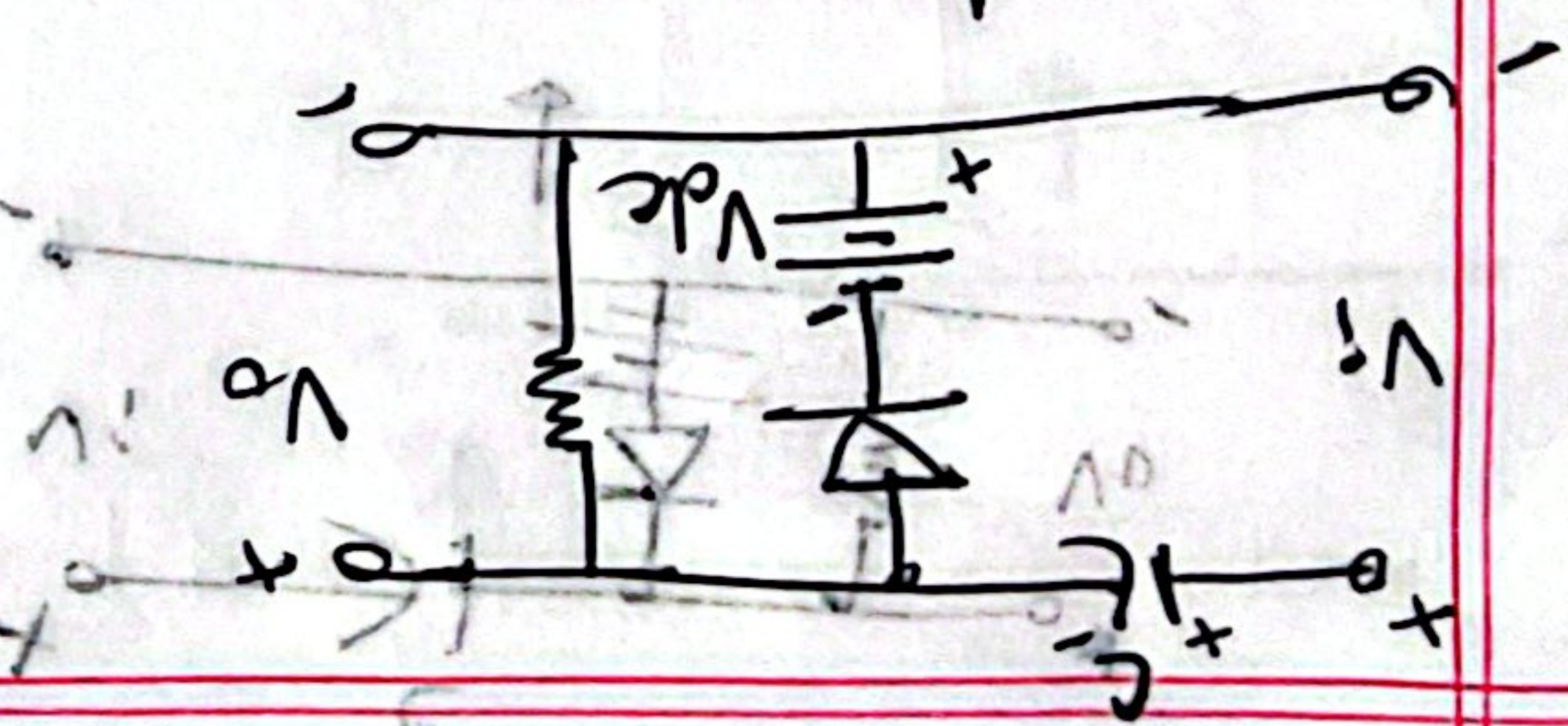
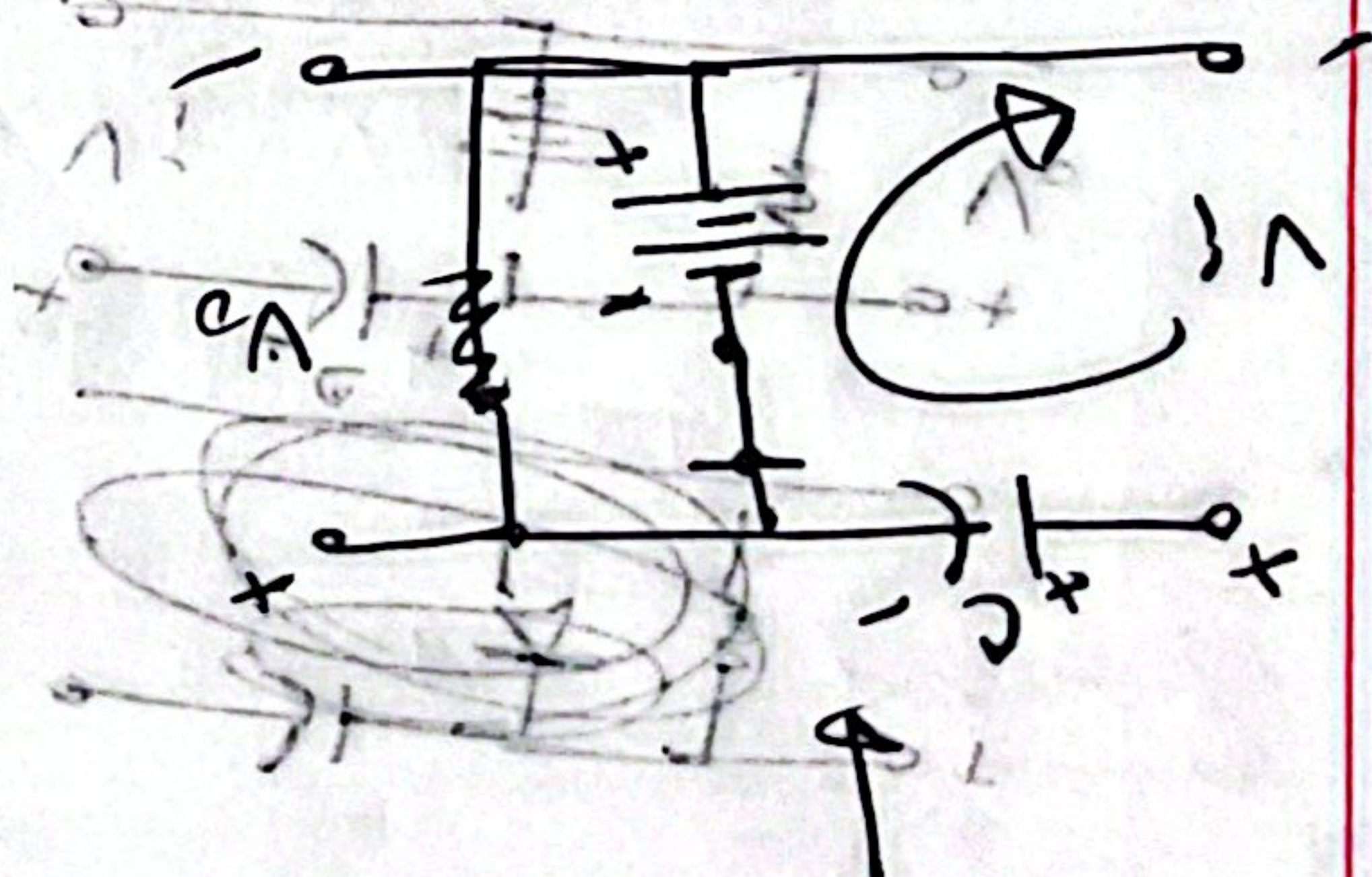
$$210 + 5$$

$$V_o = V_{dc}$$

$$-V_o + V_{dc} = 0$$

KVL in short loop

For positive half cycle



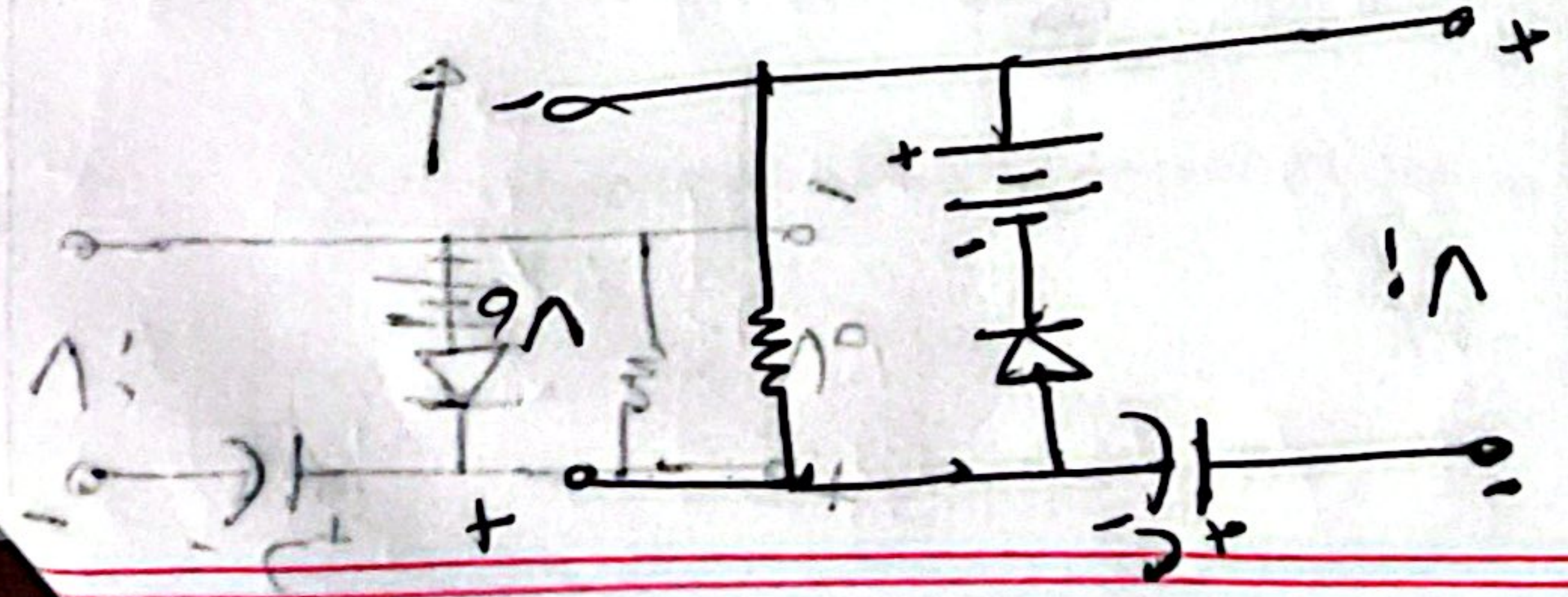
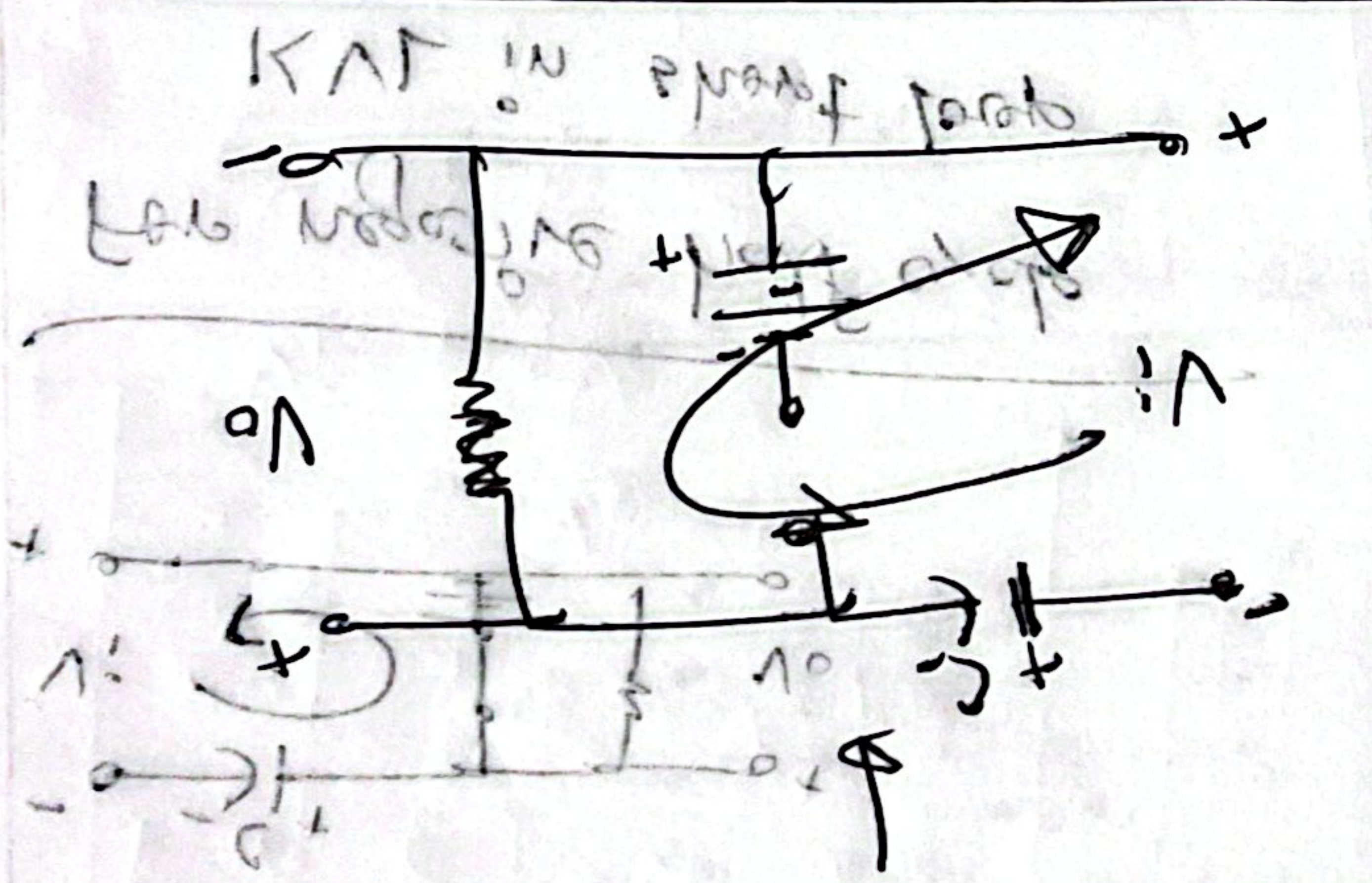
$$V_o = 21V$$

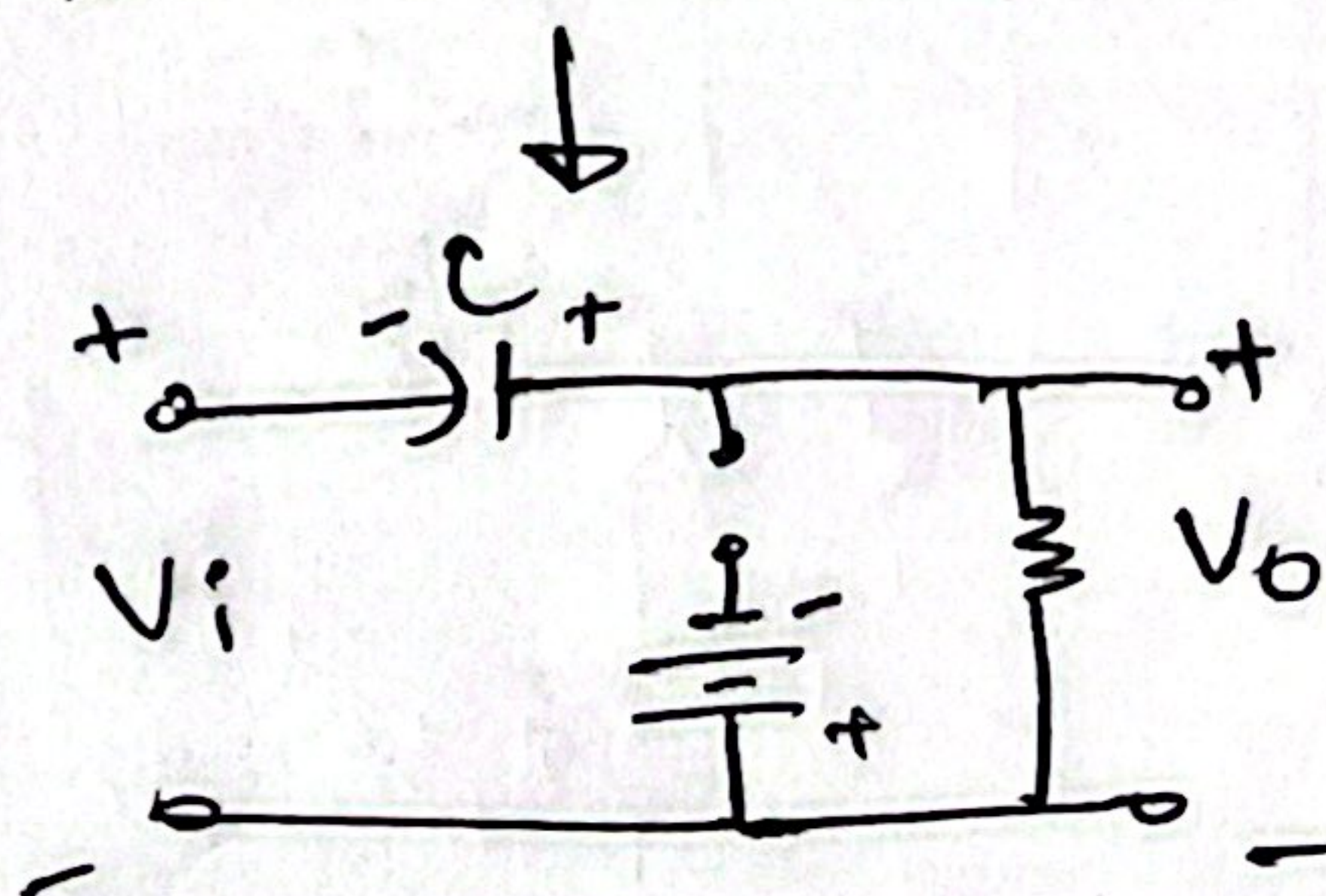
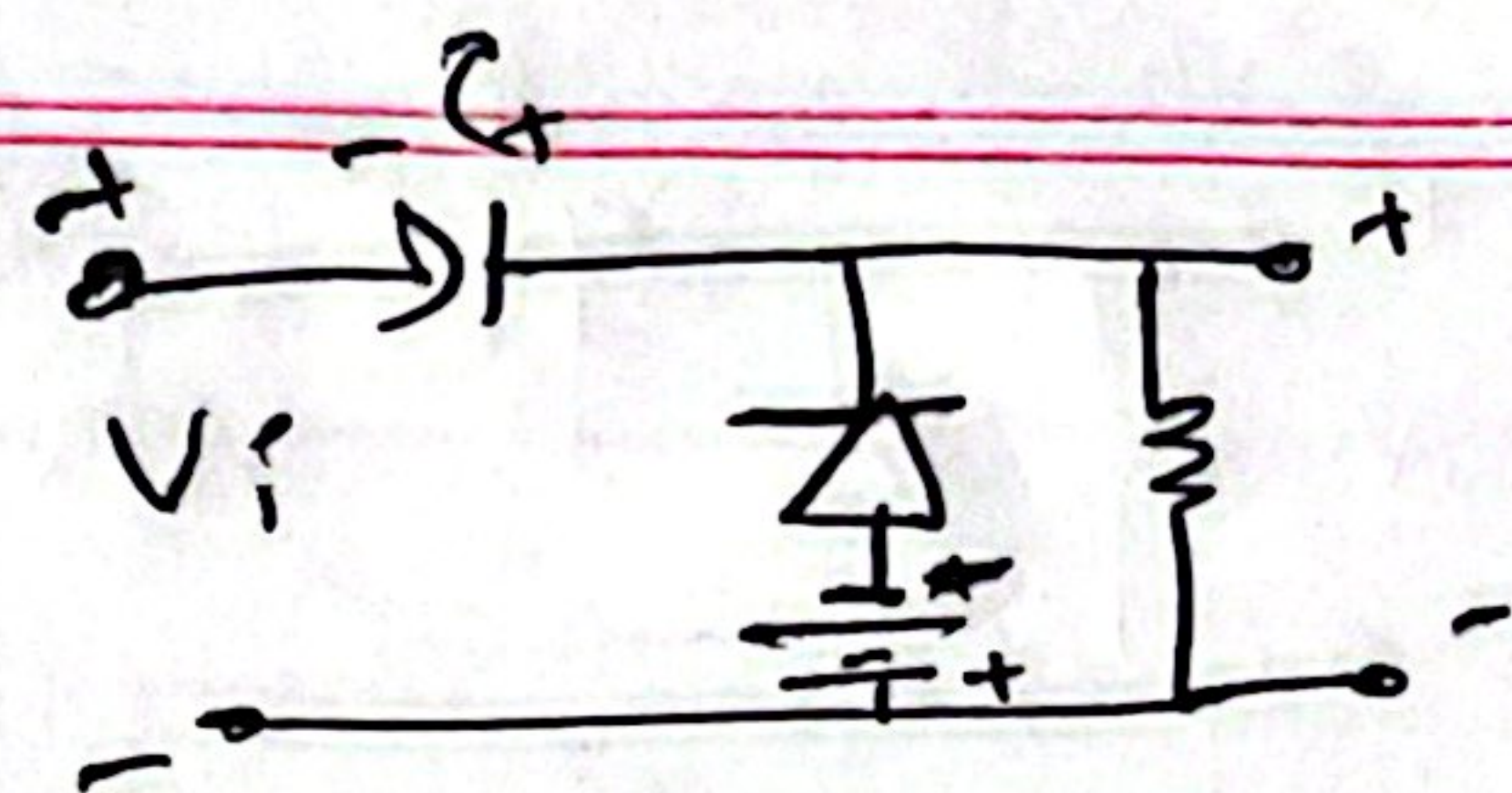
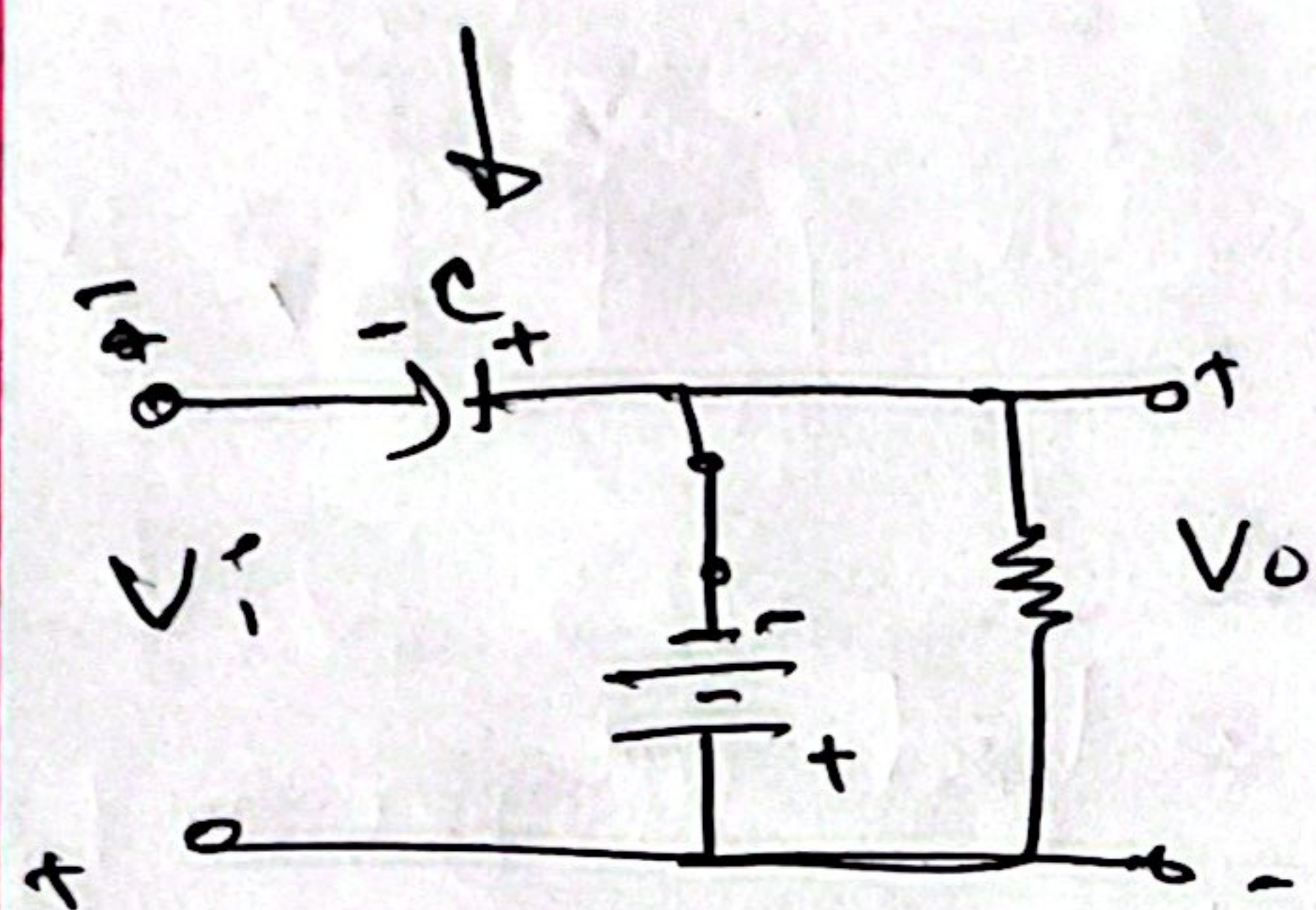
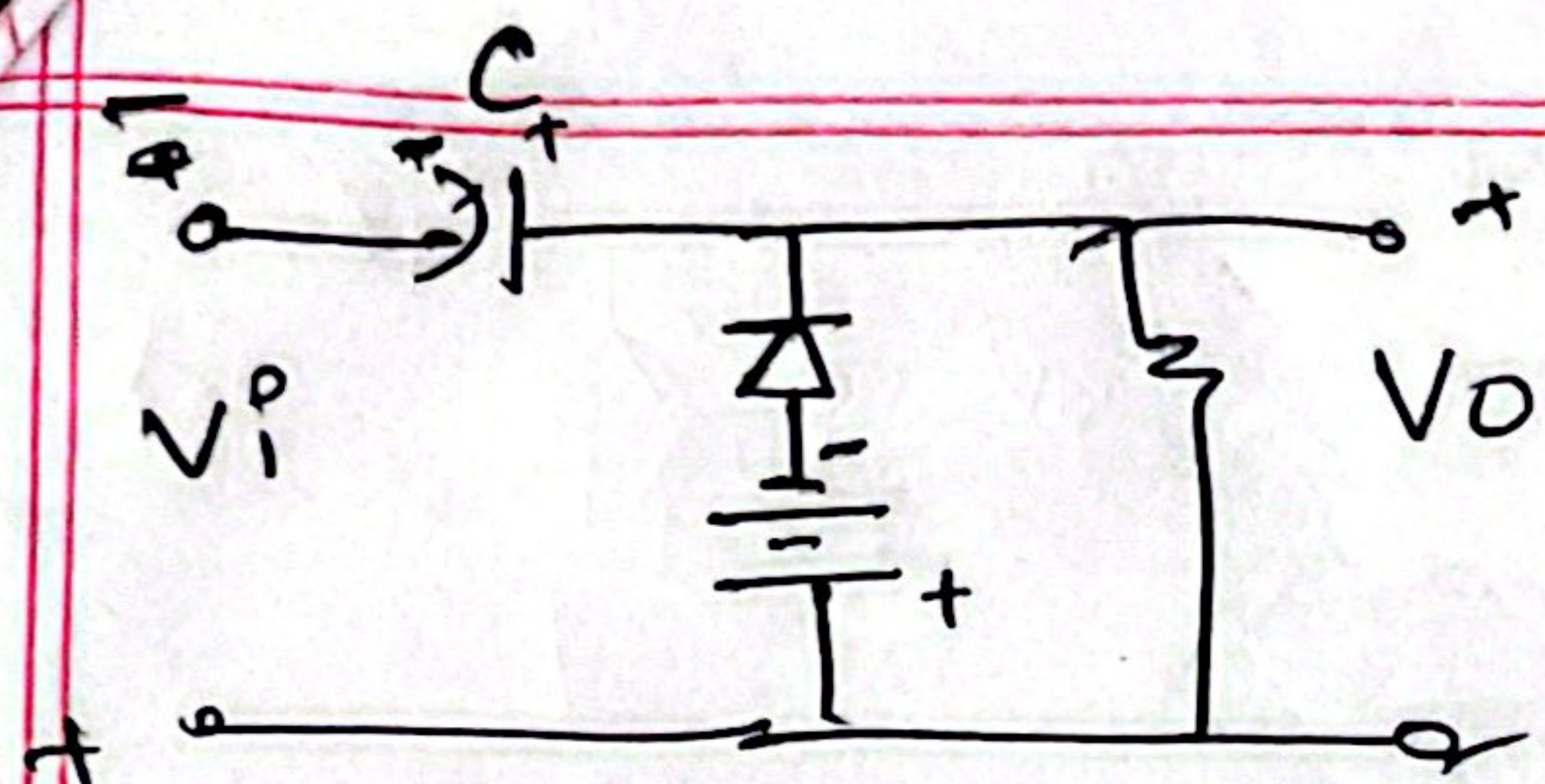
$$V_o = V_{dc}$$

$$V_o + V_{dc} = 0$$

KVL in open loop

For negative half cycle





For negative half cycle

KVL in open short loop:

$$+V_i - C - V_{dc} = 0$$

$$\begin{aligned} \Rightarrow C &= V_i - V_{dc} \\ &= 10 - 5 \\ &= 5 \text{ V} \end{aligned}$$

$$V_o = -V_{dc}$$

$$V_o = -5$$

For positive half cycle

KVL in open loop

$$-V_i - C + V_o = 0$$

$$\begin{aligned} \Rightarrow V_o &= V_i + C \\ &= 10 + 5 \\ &= 15 \end{aligned}$$

